AMRITA
VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY

User Interface Design

23CSE113

Lab Manual

Submitted by	
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Year/sem/section	1st year / 1st sem / cse-c
Credits	03

Submitted to	DR. Raj Kumar Batchu
DESIGNATION	ASST.PROFFESSOR
DEPARTMENT	SCHOOL OF COMPUTING

SN.NO	Title	Page.no	Remarks
1.	WEEK 1 : 1. Installation of VS Code tool. 2. Understanding basic of html structure. 3. Understanding basic html tags .	1-12	
2.	Week 2 : 1.Design a basic webpage with headings ,paragraph ,pre ,table by adding styles.	12-25	
3.	Week 3 : 1) create a class timetable using HTML elements. 2) Create a HTML web page that should display all subjects as an image of all syllabus.	25-35	
4.	Week4: 1)Design a registration form using only HTML tags (no CSS) with proper labels, table alignment, background color, and borders. 2)Two forms displayed side by side using iframes .one is login and another registration	35-44	
5.	Week5: 1)creating a card layout using tag consider three cards and a button using html and css . 2)navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page. 3)Design amrita login page by showing different campus on home page.upon clicking on each campus the details about each campus should be display on the same page use html and css to design a stylish webpage.	45-70	
6.	Week-6: (6)Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-	71-88	

	appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.		
7.	Week-7 : create a GitHub account and create a repository in it and push all the week 6 code files in it.	89-91	
8.	Week-8 : A) create a web page that displays a square that continuously rotates 360ing css animation. B) create a ball that moves up and down continuously using css animation. C) create a circle that moves horizontally and rotates At the same time.	91-101	
9.	Week-9: Create a webpage using html,css,js based on the give conditions, create a button for each program , based on the button, clicked the corresponding program has to be executed and should display the logics for each button includes finding an even or odd number , prime or not , factorial of a number , Armstrong or not use appropriate design to display the output.	101-111	
10.	Week-10: A) Write a java script conditional statement to find the sign of product of three numbers using functions. Display an alert box with specified sign. B) Write a java script conditional statement to sort three numbers using functions. Display an alert box to show result.	111-124	

	C) Write a java script conditional statement to find largest of five numbers using functions. Disply an alert box to show result. D) Write a java script loop that will iterate from 0 to 15 for each iteration it will check odd or even . Disply an a message to screen. E) Write a java script program to read 6 subjects marks from student then compute the average marks of student. Then this average isused to determine the corresponding grade.		
11.	Week-11: 1. A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. 2. A registration form should validate email format before submission. If invalid, the form should not be submitted. 3. A system should check password strength while typing and display whether it is weak or strong. 4. A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits. 5. Design an HTML page, such that the registration form must validate: Name (not empty) Email (valid format) Password (minimum 6 characters)	124-136	

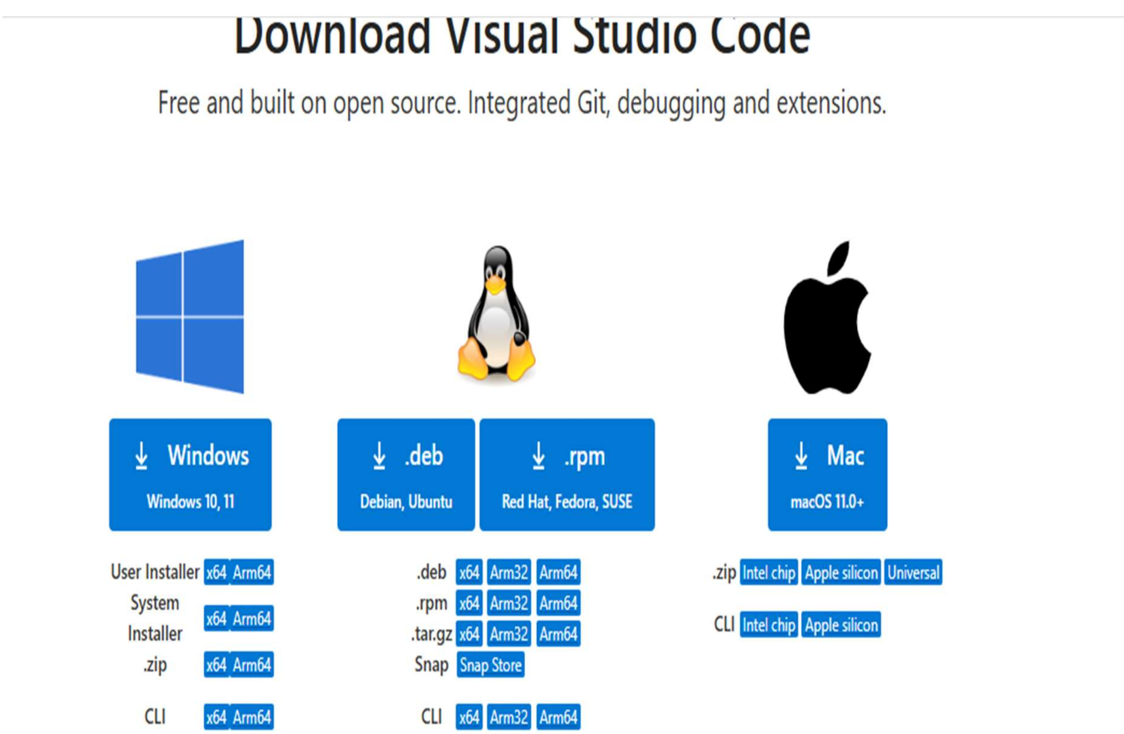
12.	WEEK-12: To install and setup the Electron Framework application.	136-145	

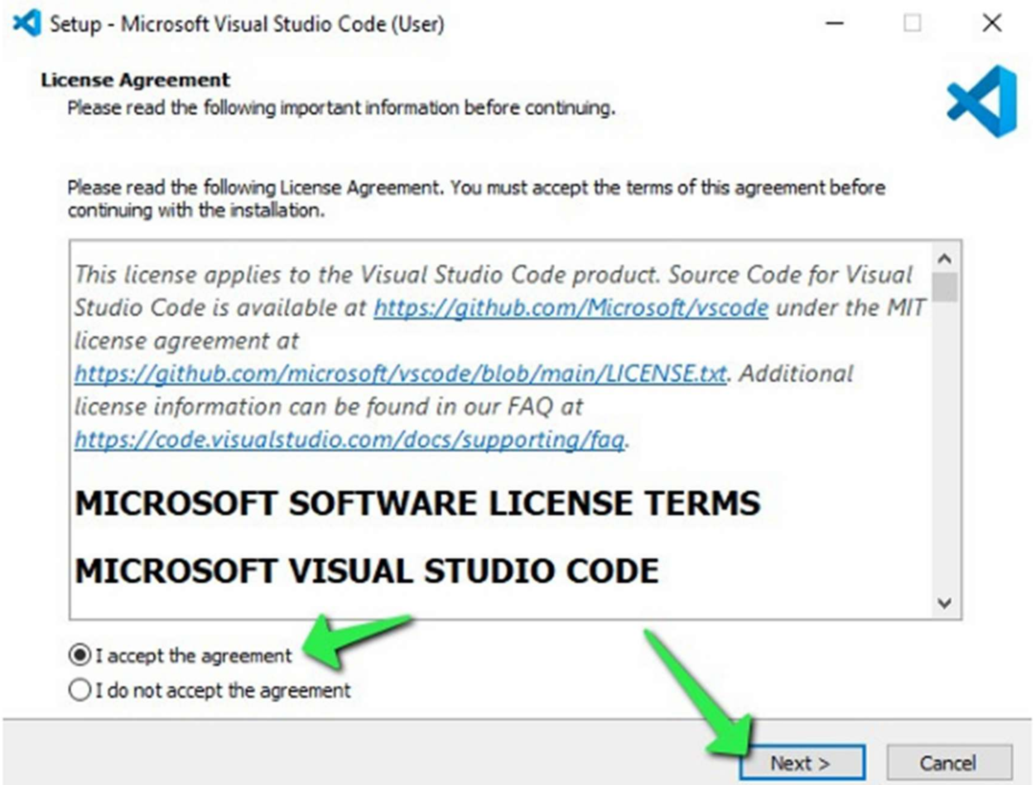
<u>WEEK -1</u>

1. Aim : Steps to install VS Code tool.

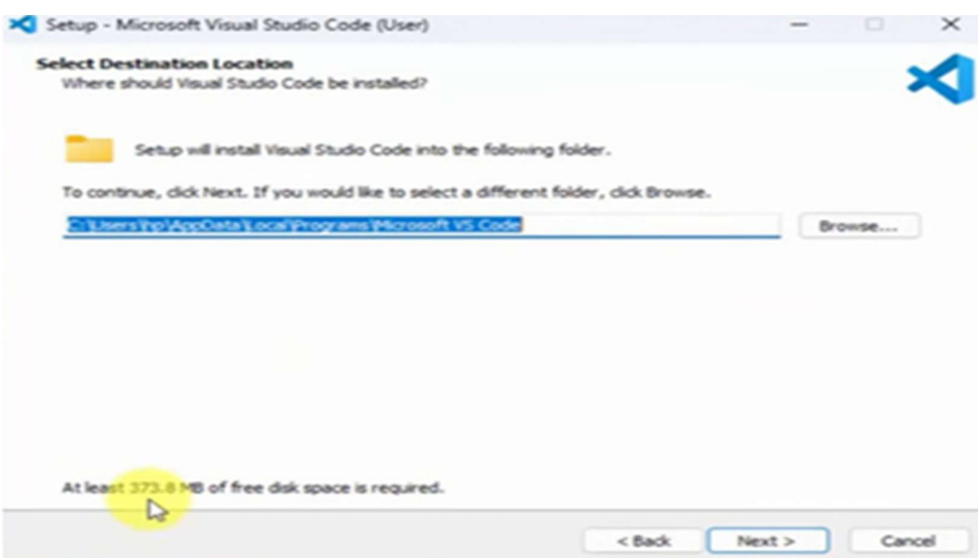
<u>Procedure:</u>

To download Visual Studio Code on any operating system, open your browser and go to **code.visualstudio.com**. The website automatically detects whether you’re using Windows, macOS, or Linux and shows the correct download option. Click the download button, run the installer once it finishes downloading, and follow the simple on-screen steps.

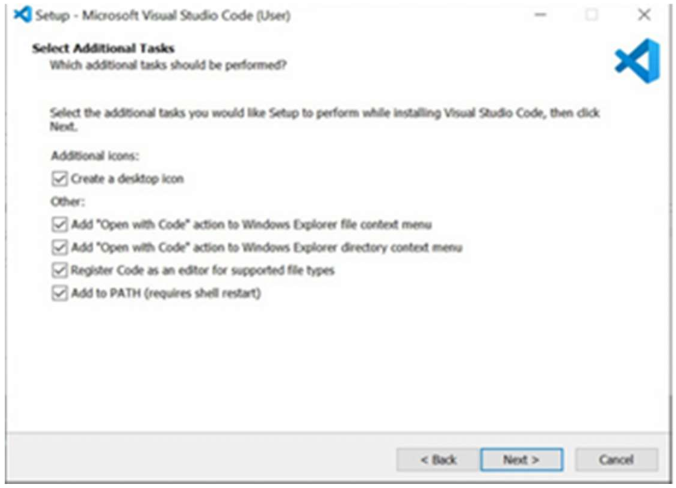


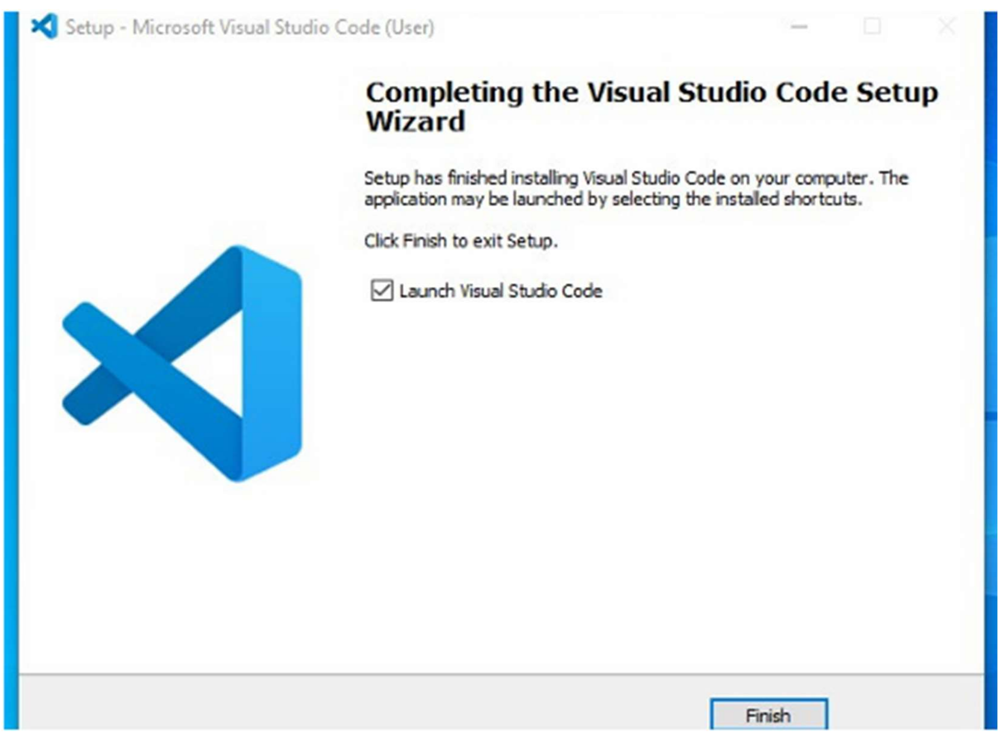


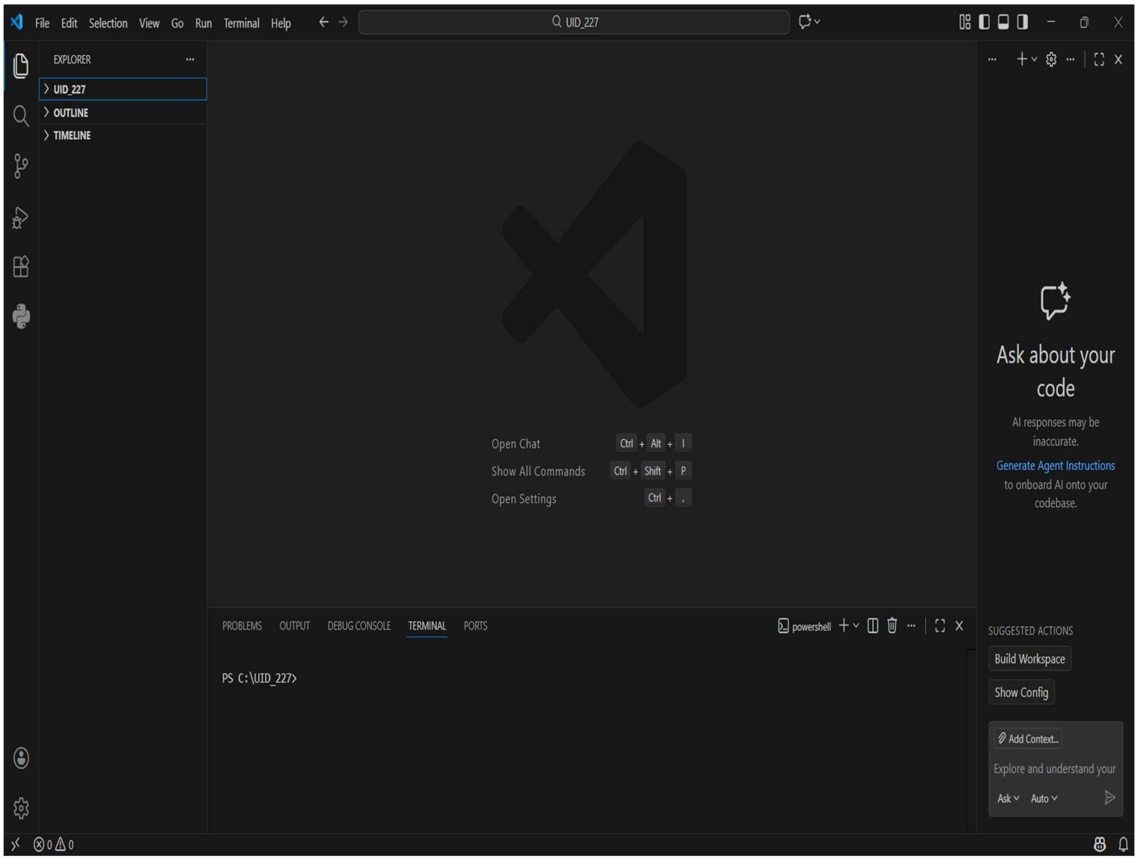
Setup - Microsoft Visual Studio Code (User)
License Agreement
Please read the following important information before continuing.
Please read the following License Agreement. You must accept the terms of this agreement before continuing with the installation.
This license applies to the Visual Studio Code product. Source Code for Visual Studio Code is available at https://github.com/Microsoft/vscode under the MIT license agreement at https://github.com/microsoft/vscode/blob/main/LICENSE.txt. Additional license information can be found in our FAQ at https://code.visualstudio.com/docs/supporting/faq.
MICROSOFT SOFTWARE LICENSE TERMS
MICROSOFT VISUAL STUDIO CODE
I accept the agreement
I do not accept the agreement
Next >
Cancel



Setup - Microsoft Visual Studio Code (User)
Select Destination Location
Where should Visual Studio Code be installed?
Setup will install Visual Studio Code into the following folder.
To continue, click Next. If you would like to select a different folder, click Browse.
Browse...
MB of free disk space is required.
< Back
Next >
Cancel


Setup - Microsoft Visual Studio Code (User)
Select Additional Tasks
Which additional tasks should be performed?
Select the additional tasks you would like Setup to perform while installing Visual Studio Code, then click Next.
Additional icons:
Create a desktop icon
Other:
Add "Open with Code" action to Windows Explorer file context menu
Add "Open with Code" action to Windows Explorer directory context menu
Register Code as an editor for supported file types
Add to PATH (requires shell restart)
< Back
Next >
Cancel


Setup - Microsoft Visual Studio Code (User)
Completing the Visual Studio Code Setup Wizard
Setup has finished installing Visual Studio Code on your computer. The application may be launched by selecting the installed shortcuts.
Click Finish to exit Setup.
Launch Visual Studio Code
Finish


File Edit Selection View Go Run Terminal Help
UID_227
EXPLORER
UID_227
OUTLINE
TIMELINE
Open Chat Ctrl + Alt + I
Show All Commands Ctrl + Shift + P
Open Settings Ctrl + ,
Ask about your code
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS C:\UID_227>
SUGGESTED ACTIONS
Build Workspace
Show Config
Add Context...
Explore and understand your
Ask Auto

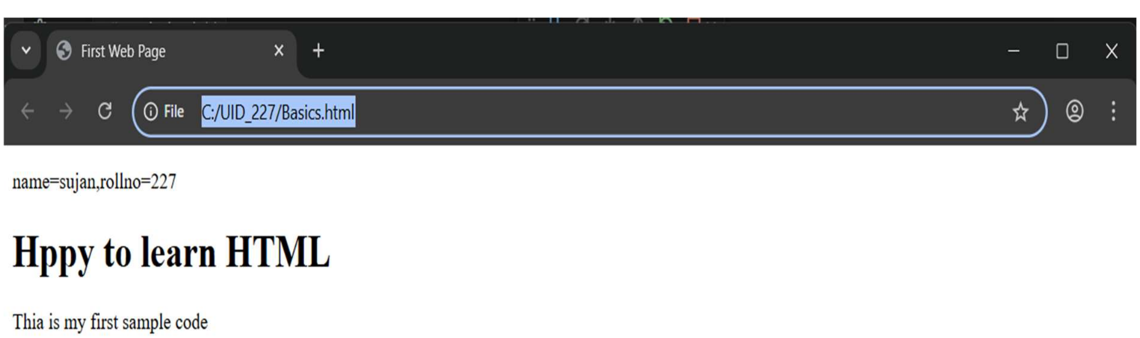
2. Aim : Understanding basics of html

Program:

UID25227-Sujan

```
<!DOCTYPE html>
<html>
    </head>
      <title>First Web Page</title>
    </head>
    <body>
      <p>name=sujan,rollno=227</p>
      <h1>Hppy to learn HTML</h1>
      <p>Thia is my first sample code</p>
      </body>
</html>
```

Output:



<u>Explanation of the tags :</u>

HTML stands for **Hyper Text MarkupLanguage**. HTML is used to **create and structure web pages**. It tells the web browser **what content to show and how to show it**.

<!DOCTYPE html>:
This declaration tells the browser that the document is written in **HTML5**.

<html>:
The <html> element is the **main/root element** of an HTML page.

<head>:
The <head> element contains **information about the webpage**, like title and meta data.

<title>:
The <title> element sets the **title of the webpage**, shown on the browser tab.

<body>:
The <body> element contains **all visible content** such as text, images, links, tables, etc.

<h1>:
The <h1> element is used to create a **heading**

<p>:
The <p> element is used to write a **paragraph**.

<html>:
This tag **starts** the HTML document.
</html>:
This tag **ends** the HTML document.
<head>:
This tag **starts** the head section which contains page information.
</head>:
This tag **ends** the head section.
<title>:
This tag **starts** the title of the webpage.
</title>:
This tag **ends** the title.
<body>:
This tag **starts** the visible content of the webpage.
</body>:
This tag **ends** the visible content.
<h1>:
This tag **starts** a heading.
</h1>:
This tag **ends** the heading.
<p>:
This tag **starts** a paragraph.
</p>:
This tag **ends** the paragraph.

3. Aim : Understanding basic html tags.

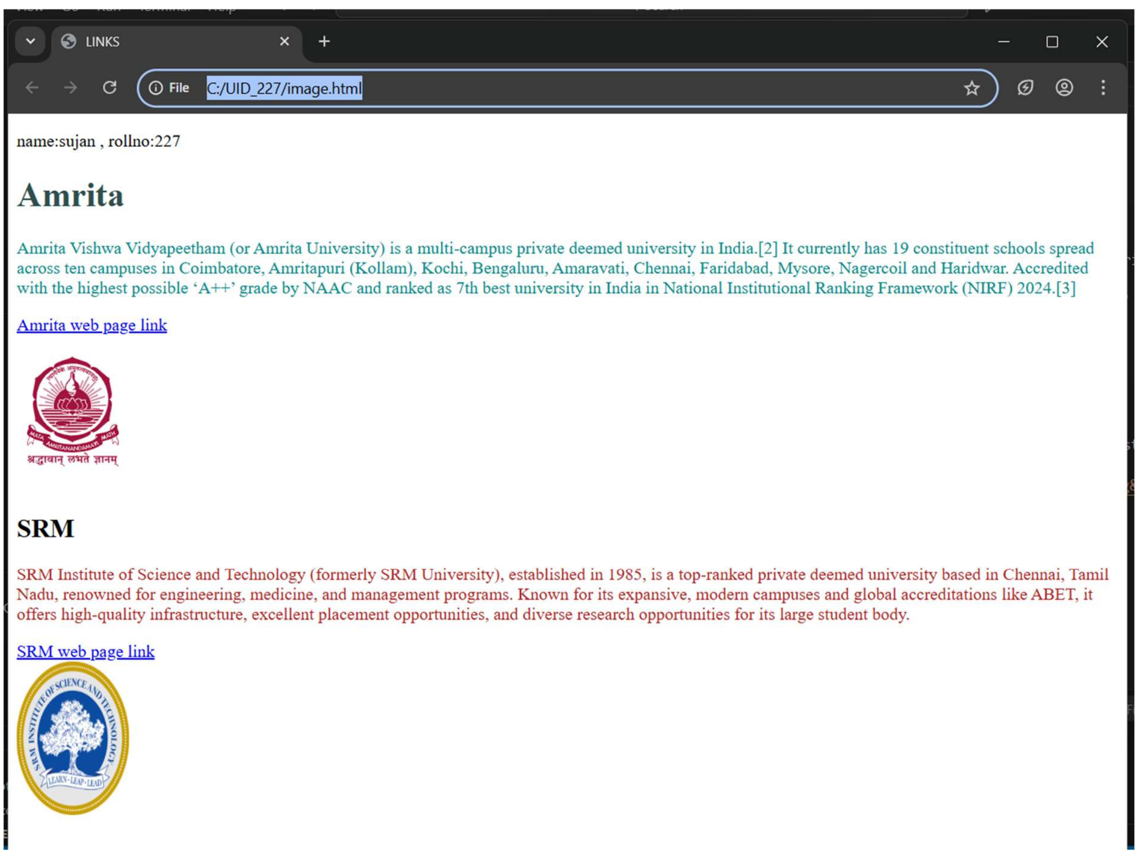
Program:

```
<!DOCTYPE html>
<html>
<head>
    <title>LINKS</title>
</head>
<body>
    <p>name:sujan , rollno:227</p>
    <h1 style="color: darkslategray;">Amrita</h1>
    <p style="color: darkcyan;">Amrita Vishwa
Vidyapeetham (or Amrita University) is a multi-
campus private deemed university in India.[2] It
currently has 19 constituent schools spread across ten
campuses in Coimbatore, Amritapuri (Kollam),
Kochi, Bengaluru, Amaravati, Chennai, Faridabad,
Mysore, Nagercoil and Haridwar. Accredited with the
highest possible ‘A++’ grade by NAAC and ranked as
7th best university in India in National Institutional
Ranking Framework (NIRF) 2024.[3]</p>
    <a href="https://www.amrita.edu/">Amrita web
page link</a><br>
```

```
<img
src="https://upload.wikimedia.org/wikipedia/en/3/30/
Amrita_Vishwa_Vidyapeetham_-_Logo_Icon.svg"
        alt="Amrita University Logo"
        width="104"
        height="142">



<h2 style="color: black;">SRM</h2>
<p style="color: firebrick;">SRM Institute of
Science and Technology (formerly SRM University),
established in 1985, is a top-ranked private deemed
university based in Chennai, Tamil Nadu, renowned
for engineering, medicine, and management
programs. Known for its expansive, modern
campuses and global accreditations like ABET, it
offers high-quality infrastructure, excellent placement
opportunities, and diverse research opportunities for
its large student body. </p>
<a href="https://www.srmist.edu.in/">SRM web
page link</a><br>
<img src="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcQYihGewCy
Haa81vhA7V_Oesqtqz5W08Rfo3g&s"
```

```
        alt="SRM University Logo"
        width="104"
        height="142">

</body>
</html>
```

Output:



Explanation:

<!DOCTYPE html> :
It tells the browser that this document is written in HTML5. It helps the browser render the page correctly.

<html> :
This is the root element of the HTML document. All other tags are written inside this tag.

<head> :
Used to store metadata of the webpage such as the title. Content inside this tag is not visible on the webpage.

<title> :
Defines the title of the webpage, which appears on the browser tab.

<body> :
Contains all the visible content of the webpage like headings, paragraphs, links, and images.

<h1> and <h2> :
Used for headings.
<h1> represents the main heading and <h2> represents a sub-heading.
The style attribute is used here to apply color to the headings.

<p> :
Used to display text in paragraph format.
The style attribute is used to change the text color.

<a href="URL"> :
The anchor tag is used to create hyperlinks.
href specifies the destination URL and the text inside the tag becomes clickable.

 :
Used to insert a line break and move content to the next line.
It does not have a closing tag.

<img> :
Used to display images on the webpage.
It does not require a closing tag.

- src → specifies the image URL
- alt → alternate text if the image fails to load (important for accessibility)
- width → image width in pixels
- height → image height in pixels

style attribute :
Used to apply inline CSS styling such as text color inside tags like <h> and <p>.

Week-2

1. AIM : Design a basic webpage with headings ,paragraph ,pre ,table by

adding styles.

CODE:

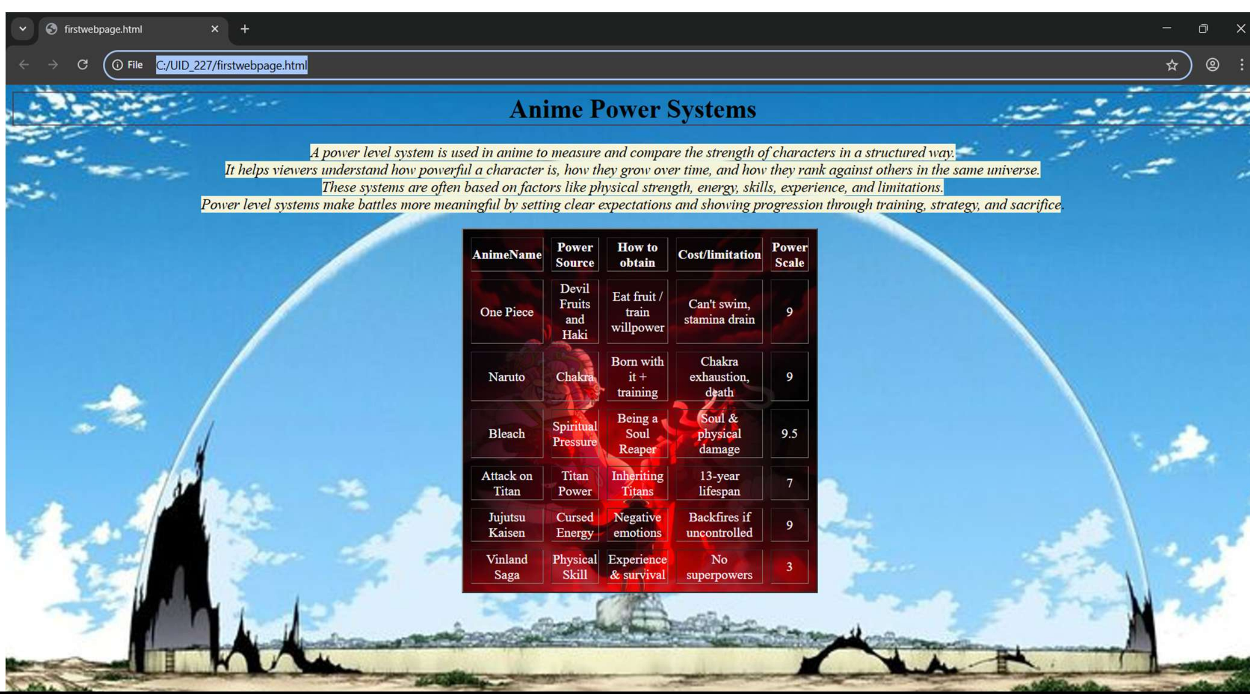
```
<html>
  <body style="background-image:url(soulsocity.jpg);background-size: cover;
background-color: rgb(31, 9, 9);">
     <h1 style="text-align: center;color: #030303;
;border: 2px solid rgba(81, 36, 36, 0.6);">Anime Power Systems</h1>
     <p style="color: rgb(12, 12, 12);text-align:center;font-size:large;"><i><mark style="background-color: beige;">A power level system is used in anime to measure and compare the strength of characters in a structured way.<br> It helps viewers understand how powerful a character is, how they grow over time, and how they rank against others in the same universe.<br> These systems are often based on factors like physical strength, energy, skills, experience, and limitations.<br> Power level systems make
```

```
battles more meaningful by setting clear expectations and
showing progression through training, strategy, and
sacrifice</mark></i>.</p>

    <table width="420" border="2" cellspacing="10" style="text-
align: center;color: rgb(253, 245, 250) ;margin-left:
550px;background-size:cover;background-
image:url(iconicwalk.jpg)">

      <tr>

        <th>AnimeName</th>

        <th>Power Source </th>

        <th>How to obtain</th>

        <th>Cost/limitation</th>

        <th>Power Scale</th>

      </tr>

      <tr>

        <td>One Piece</td>

        <td>Devil Fruits and Haki</td>

        <td>Eat fruit / train willpower</td>

        <td>Can't swim, stamina drain</td>

        <td>9</td>

      </tr>

      <tr>

        <td>Naruto</td>

        <td>Chakra</td>

        <td>Born with it + training</td>
```

```
    <td>Chakra exhaustion, death</td>
    <td>9</td>
</tr>
<tr>
    <td>Bleach</td>
    <td>Spiritual Pressure</td>
    <td>Being a Soul Reaper</td>
    <td>Soul & physical damage</td>
    <td>9.5</td>
</tr>
<tr>
    <td>Attack on Titan</td>
    <td>Titan Power</td>
    <td>Inheriting Titans</td>
    <td>13-year lifespan</td>
    <td>7</td>
</tr>
<tr>
    <td>Jujutsu Kaisen</td>
    <td>Cursed Energy</td>
    <td>Negative emotions</td>
    <td>Backfires if uncontrolled</td>
    <td>9</td>
</tr>
```

```
<tr>
   <td>Vinland Saga</td>
   <td>Physical Skill</td>
   <td>Experience & survival</td>
   <td>No superpowers</td>
   <td>3</td>
</tr>
</table>


</body>


</html>
```

OUTPUT:



Explanation: (Contains only new tags I used)

1. <table> = the <table> tag is used to arrange data into rows and columns.

2. <tr> = Creates a horizontal row.

3. <th> = Defines a header cell (usually bold and centred by default).

4. <td> = the <td> tag stands for table Data.

5. colspan attribute = It is used to merge multiple columns across horizontally.

6. rowspan attribute = It is used to merge multiple rows across vertically.

Week-3

2. **Aim :** a) create a class timetable using HTML elements.

b) Create a HTML web page that should display all subjects as an image of all syllabus.

Code:

```
<html>
  <body>
    <title>Time Table</title>
    <p>Name:Sujan,Rollno:25227,sec:cse-c</p>

    <table border="3" cellspacing="2" cellpadding="6"
width="100%" style="border-collapse: collapse; text-align:
center;">
      <tr>
        <td colspan="6" style="text-align: left;">
          <b>Dept:</b> CSE<br>
```

```
      <b>Class:</b> B.Tech<br>
      <b>sem:</b> 2 <br>
    </td>
    <td colspan="6" style="text-align: left;">
      <b>class room </b> 512
    </td>
  </tr>
  <tr>
    <td colspan="6" style="text-align: left;">
      <b>Class Advisors:</b><br>
      <b>1. Name: MR.BJNL</b><br>
      <b>Email:bj_lashminarayana@av.amrita.edu</b><br>
      <b>Mobile Number : +91- 7989795768<b><br>
    </td>
    <td colspan="6" style="text-align: left;">
      <b>Class Advisors:</b><br>
      <b>1. Name:MR.Arun Kumar</b><br>
      <b>Email:s_arunkumar@av.amrita.edu</b><br>
      <b>Mobile Number:+91-89408572202<b><br>
    </td>
  </tr>
  <tr>
    <th rowspan="2">Time/Day</th>
    <th>Slot-1</th>
```

```
  <th>Slot-2</th>
  <th rowspan="2">10:35<br>10:45</th>
  <th>Slot-3</th>
  <th>Slot-4</th>
  <th>Slot-5</th>
  <th rowspan="2">1:15<br>2:05</th>
  <th>Slot-6</th>
  <th rowspan="2">3:00t<br>3:10</th>
  <th>Slot-7</th>
  <th>Slot-8</th>
</tr>
<tr>
  <th>8:55-9:45</th>
  <th>9:45-10:35</th>
  <th>10:45-11:35</th>
  <th>11:35-12:25</th>
  <th>12:25-1:15</th>
  <th>2:05-3:00</th>
  <th>3:10-4:00</th>
  <th>4:00-4:50</th>
</tr>
<tr>
  <th>Monday</th>
```

```
        <th colspan="2">23MAT116 Discrete Mathematics_lab2</th>
        <th rowspan="5" style="background-color: yellow;">S<br>h<br>o<br>t<br><br>B<br>r<br>e<br>a<br>k</th>
        <th>23PHY115 Modern<br>Physics</th>
        <th>23CSE111ObjectOriented<br>Programming</th>
        <th>22ADM111Glimpses of<br>Glorious India</th>
        <th rowspan="5" style="background-color: yellow;">L<br>u<br>n<br>c<br>h<br>B<br>r<br>e<br>a<br>k</br>a</th>
        <th>22ADM111Glimpses of<br>Glorious India(P)</th>
        <th rowspan="5" style="background-color: yellow;">S<br>h<br>o<br>t<br><br>B<br>r<br>e<br>a<br>k</th>
        <th colspan="2">Library/Sports</th>
    </tr>
    <tr>
        <th>Tuesday</th>
        <th colspan="2">23CSE113 User Interface Design LAB(BYOD)</th>
        <th>23MAT117 Linear<br>Algebra</th>
        <th>23MAT116 Discrete<br>Mathematics</th>
        <th style="color: red;">COUNSELLING</th>
        <th>23MEE115 Manufacturing<br>Practice LAB</th>
```

```
<th colspan="2">23MEE115 Manufacturing<br>Practice LAB</th>
</tr>
<tr>
<th>Wednesday</th>
<th>23PHY115 Modern<br>Physics</th>
<th>23CSE113 User Interface Design</th>
<th colspan="2">23MAT117 Linear Algebra_LAB(BYOD)</th>
<th style="color: red;">COUNSELLING</th>
<th>LIBRARY</th>
<th>22ADM111Glimpses of<br>Glorious India</th>
<th>Library/Sports</th>
</tr>
<tr>
<th>Thursday</th>
<th>23CSE111ObjectOriented<br>Programming</th>
<th>23MAT116 Discrete<br>Mathematics</th>
<th colspan="2">
23CSE111Object Oriented Programming <br>_Lab2
</th>
<th>23MAT117<br>LinearAlgebra</th>
<th>LIBRARY</th>
<th colspan="2">Library/Sports</th>
```

```
        </tr>
        <tr>
          <th>Friday</th>
          <th>23CSE111ObjectOriented<br>Programming</th>
          <th>23MAT116 Discrete<br>Mathematics</th>
          <th>23PHY115 Modern<br>Physics</th>
          <th>23CSE113 User Interface Design</th>
          <th>23MAT117<br>LinearAlgebra</th>
          <th>LIBRARY</th>
          <th colspan="2">Library/Sports</th>
        </tr>
      </table>

      <br><br>

      <h3 style="text-align:center;">Theory & Lab</h3>
      <table border="1" cellspacing="2" cellpadding="6"
width="100%" style="border-collapse:collapse;">
        <tr>
          <th>Course Code</th>
          <th>Course Name</th>
          <th>L-T-P</th>
          <th>Credits</th>
          <th>Faculty Name</th>
```

```html
    <th>Assisting Faculty</th>
`   </tr>
<tr>
    <td>23MAT116</td>
    <td>Discrete Mathematics</td>
    <td>3-0-2</td>
    <td>4</td>
    <td>Dr. Srivivasa Rao Thota</td>
    <td></td>
</tr>
<tr>
    <td>23MAT117</td>
    <td>Linear Algebra</td>
    <td>3-0-2</td>
    <td>4</td>
    <td>Mr. Sapavat Bixapathi</td>
    <td></td>
</tr>
<tr>
    <td>23CSE111</td>
    <td>Object Oriented Programming</td>
    <td>3-0-2</td>
    <td>4</td>
    <td>Dr. Satya Rajendra Singh R</td>
```

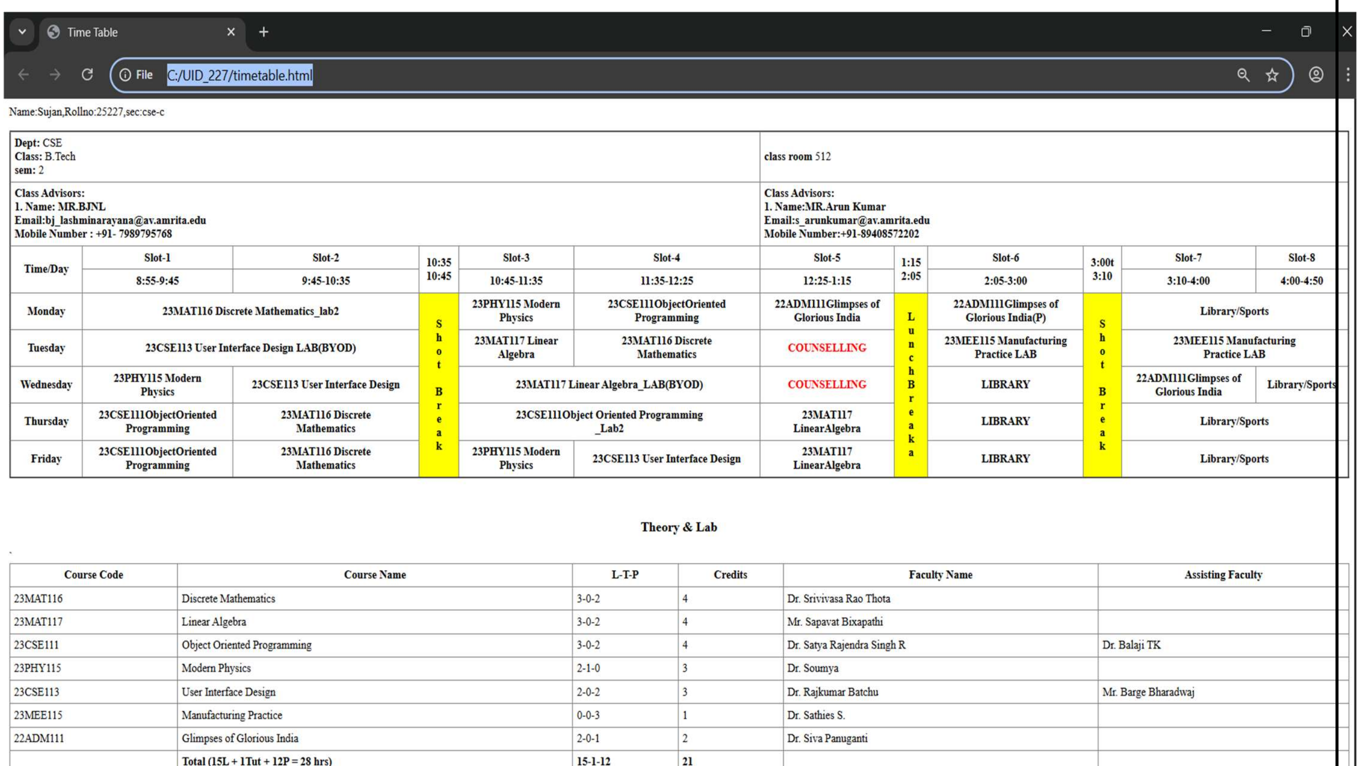
```
    <td>Dr. Balaji TK</td>
</tr>
<tr>
    <td>23PHY115</td>
    <td>Modern Physics</td>
    <td>2-1-0</td>
    <td>3</td>
    <td>Dr. Soumya</td>
    <td></td>
</tr>
<tr>
    <td>23CSE113</td>
    <td>User Interface Design</td>
    <td>2-0-2</td>
    <td>3</td>
    <td>Dr. Rajkumar Batchu</td>
    <td>Mr. Barge Bharadwaj</td>
</tr>
<tr>
    <td>23MEE115</td>
    <td>Manufacturing Practice</td>
    <td>0-0-3</td>
    <td>1</td>
    <td>Dr. Sathies S.</td>
```

```
      <td></td>
    </tr>
    <tr>
      <td>22ADM111</td>
      <td>Glimpses of Glorious India</td>
      <td>2-0-1</td>
      <td>2</td>
      <td>Dr. Siva Panuganti</td>
      <td></td>
    </tr>
    <tr>
      <td></td>
      <td><b>Total (15L + 1Tut + 12P = 28 hrs)</b></td>
      <td><b>15-1-12</b></td>
      <td><b>21</b></td>
      <td></td>
      <td></td>
    </tr>
  </table>
  <br>
  <br>
  <br>
  <br>
```

```
<h2 style="text-align: center;">Syllabus</h2>
<h3>Discert Maths</h3>
<img
src="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 221641.png" width="1000" height="400">
<br>
<h3>Linear Algebra</h3>
<img
src="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 222317.png" width="1000" height="400">
<br>
<h3>Object Oriented Programming</h3>
<img
src="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 222525.png" width="1000" height="400">
<br>
<h3>Modern Physics</h3>
<img
src="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 222818.png0" width="1000" height="400">
<br>
<h3>User Interface Design</h3>
<img
scr="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 223014.png" width="1000" height="400">
<br>
<h3>Manufacturing Practice</h3>
```

```
    <img
src="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 223158.png" width="1000" height="400">

    <br>

    <h3>Glimpses Of Glorious India</h3>

    <img
scr="c:\Users\sujan\OneDrive\Pictures\Screenshots\Screenshot
2026-02-04 223320.png" width="1000" height="400">

  </body>

</html>
```

Output:



Name:Sujan,Rollno:25227,sec:cse-c

Dept: CSE
Class: B.Tech
sem: 2

class room 512

Class Advisors:
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Time/Day	Slot-1	Slot-2	10:35 10:45	Slot-3	Slot-4	Slot-5	1:15 2:05	Slot-6	3:00t 3:10	Slot-7	Slot-8
	8:55-9:45	9:45-10:35		10:45-11:35	11:35-12:25	12:25-1:15		2:05-3:00		3:10-4:00	4:00-4:50
Monday	23MAT116 Discrete Mathematics_lab2		Shot Break	23PHY115 Modern Physics	23CSE111ObjectOriented Programming	22ADM111Glimpses of Glorious India	LunchBreaka	22ADM111Glimpses of Glorious India(P)	Shot Break	Library/Sports	
Tuesday	23CSE113 User Interface Design LAB(BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB	
Wednesday	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra_LAB(BYOD)		COUNSELLING		LIBRARY		22ADM111Glimpses of Glorious India	Library/Sports
Thursday	23CSE111ObjectOriented Programming	23MAT116 Discrete Mathematics		23CSE111Object Oriented Programming _Lab2		23MAT117 LinearAlgebra		LIBRARY		Library/Sports	
Friday	23CSE111ObjectOriented Programming	23MAT116 Discrete Mathematics		23PHY115 Modern Physics	23CSE113 User Interface Design	23MAT117 LinearAlgebra		LIBRARY		Library/Sports	

Theory & Lab

Course Code	Course Name	L-T-P	Credits	Faculty Name	Assisting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr. Srivivasa Rao Thota	
23MAT117	Linear Algebra	3-0-2	4	Mr. Sapavat Bixapathi	
23CSE111	Object Oriented Programming	3-0-2	4	Dr. Satya Rajendra Singh R	Dr. Balaji TK
23PHY115	Modern Physics	2-1-0	3	Dr. Soumya	
23CSE113	User Interface Design	2-0-2	3	Dr. Rajkumar Batchu	Mr. Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	1	Dr. Sathies S.	
22ADM111	Glimpses of Glorious India	2-0-1	2	Dr. Siva Panuganti	
	Total (15L + 1Tut + 12P = 28 hrs)	**15-1-12**	**21**		

Syllabus

Discert Maths

Syllabus

Unit 1
Logic, Mathematical Reasoning and Counting: Logic, Prepositional Equivalence, Predicate and Quantifiers, Theorem Proving. Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2
Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.
Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3
Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem. Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

Lab Practice Problems: Verifications of logical statements, truth table, tautology. Recursive algorithms. Graph problems, degree, shortest path algorithm, Euler's algorithm and closest neighbour algorithm.

Textbook(s)
Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw- Hill Publishing Company Limited, New Delhi, Sixth Edition, 2007.
James Strayer, "Elementary Number Theory", Waveland Press, 2002.

Linear Algebra

Syllabus

Unit 1
Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis – Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2
Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations. Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3
Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

Textbook(s)

Howard Anton and Chris Rorres, "Elementary Linear Algebra", Tenth Edition, John Wiley & Sons, 2010.

Object Oriented Programming

Syllabus
Unit1
Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system – UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram – Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit2
Aggregation and Composition using Class Diagram – Generalization using Class Diagram – Inheritance - `Constructor and Over Riding – Visibility – Attribute – Parameter – Package - Local and Global - Polymorphism – Overloading - Abstract Classes and Interfaces.

Unit3
Exception Handling - Inner Classes - Wrapper classes – String - and String Builder classes – Number – Math – Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and Arraylist - Iterator and Iterable.

Textbook(s)
Y.Daniel Liang, "Introduction to Java Programming", Tenth Edition, PHI, 2013.
Grady Booch and Robert A. Maksimchuk, "Object-oriented Analysis and Design with Applications", Third Edition, Pearson Education, 2009.

Modern Physics

Unit 1
Origin of quantum theory of radiation: Black body radiation, photo-electric effect, Compton Effect – pair production and annihilation, De-Broglie hypothesis, description of waves and wave packets,group velocities. Evidence for wave nature of particles: Davisson-Germer experiment, Heisenberguncertainty principle.

Unit 2
Atomic structure: Historical Development of atomic structures: Thomson's Model, Rutherford's Model: Scattering formula and its predictions, Atomic spectra - Bohr's Model, Sommerfield's Model, The correspondence principle, nuclear motion, and atomic excitation, Application: Lasers.

Unit 3
Quantum mechanics: Wave function, Probability density, expectation values - Schrodinger equation – time dependent and independent, Linearity and superposition, expectation values, operators, Eigen functions and Eigen values

Unit 4
Application of 1D Schrodinger Wave equation: Free particle, Particle in a box, Finite potential well,Tunnel effect, Harmonic oscillator.

Unit 5
Intro to Quantum computing- Q bits- II Quantum correlations: Bell inequalities and entanglement,Schmidt decomposition, super dense coding, teleportation. Module

User Interface Design

Syllabus

Unit 1
Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media - Web Based Forms – HTML Forms - CSS for Presentation - Basic Style Sheet - A CSS Style Primer - Using Style Classes - Using Style IDs - Formatting Text - Advanced Typography with CSS3 - Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit 2
CSS Box Model and Positioning - Creating Layouts Using Modern CSS - Backgrounds and Borders - CSS Transformations and Transitions - Animating with CSS and the Canvas - Dynamic Web Pages – Web Scripting - JavaScript programming for Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validation - Document Object Model - DOM manipulation and events - Event Handlers

Unit 3
Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility - Introduction to Electron - Overview of Electron - Setting up an Electron project - Creating a basic desktop application with Electron - Object communication - Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).

Manufacturing Practice

Syllabus

1. **Additive Manufacturing Laboratory –12 hours**

Introduction to digital manufacturing. Introduction to Additive Manufacturing - types – additive manufacturing applications - Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation- G code generation - 3D printing of simple geometries.

2. **Mechanical Engineering Laboratory –12 hours**

Study of tools and equipment used for basic manufacturing processes.
Manual arc welding practice for making Butt and Lap joints - Soldering Practice
Introduction to Machine Tools and Machining Processes

3. **Automation lab –12 hours**

Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.

4. **Automobile Engineering lab –9 hours**

Overview of automobiles – components –functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

Glimpses Of Glorious India

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

Textbook(s)
"Glimpses of Glorious India", In house publication (In print).

Explanation: (Contains only new tags I used)

1. <table> = the <table> tag is used to arrange data into rows and columns.

2. <tr> = Creates a horizontal row.

3. <th> = Defines a header cell (usually bold and centred by default).

4. <td> = the <td> tag stands for table Data.

5. colspan attribute = It is used to merge multiple columns across horizontally.

6. rowspan attribute = It is used to merge multiple rows across vertically.

Week-4

1. Aim: Creating a collage registration form using HTML.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>College Admission Registration</title>
</head>
<body bgcolor="#E6E6FA"> <table border="1" align="center"
cellpadding="10" cellspacing="0" bgcolor="white">
    <tr>
      <th colspan="2">
        <h2>College Admission Registration</h2>
      </th>
    </tr>

    <form action="#" method="post">
      <tr>
        <td><label for="name">Full Name:</label></td>
        <td><input type="text" id="name" name="name"></td>
      </tr>

      <tr>
        <td><label for="email">Email:</label></td>
```

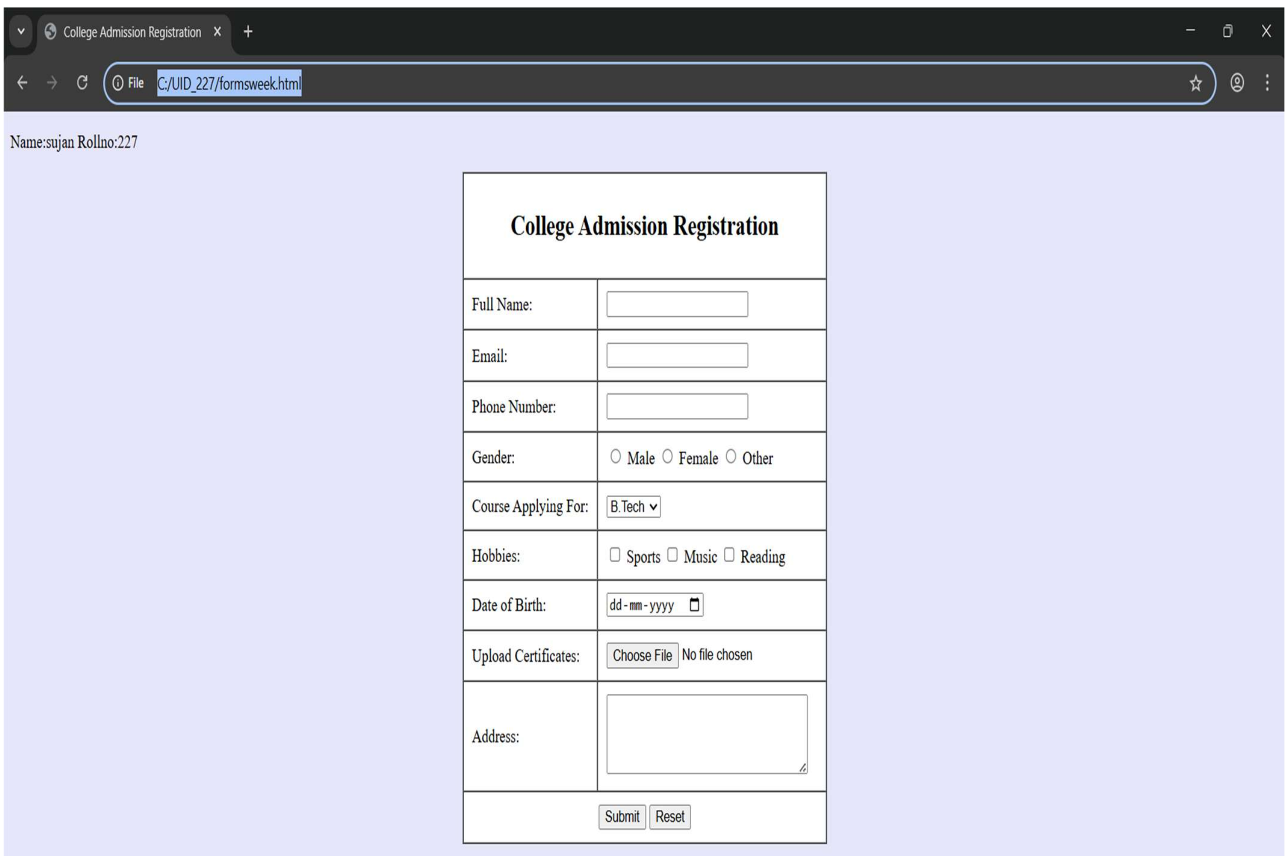
```
    <td><input type="email" id="email" name="email"></td>
  </tr>

  <tr>
    <td><label for="phone">Phone Number:</label></td>
    <td><input type="tel" id="phone" name="phone"></td>
  </tr>

  <tr>
    <td>Gender:</td>
    <td>
      <input type="radio" id="male" name="gender"
value="male">
      <label for="male">Male</label>
      <input type="radio" id="female" name="gender"
value="female">
      <label for="female">Female</label>
      <input type="radio" id="other" name="gender"
value="other">
      <label for="other">Other</label>
    </td>
  </tr>

  <tr>
    <td>Course Applying For:</td>
```

```
        <td>
          <select name="course">
            <option value="btech">B.Tech</option>
            <option value="mtech">M.Tech</option>
            <option value="bsc">B.Sc</option>
          </select>
        </td>
      </tr>

      <tr>
        <td>Hobbies:</td>
        <td>
          <input type="checkbox" id="sports" name="hobbies"
value="sports">
          <label for="sports">Sports</label>
          <input type="checkbox" id="music" name="hobbies"
value="music">
          <label for="music">Music</label>
          <input type="checkbox" id="reading" name="hobbies"
value="reading">
          <label for="reading">Reading</label>
        </td>
      </tr>

      <tr>
```

```
            <td>Date of Birth:</td>
            <td><input type="date" name="dob"></td>
          </tr>

          <tr>
            <td>Upload Certificates:</td>
            <td><input type="file" name="certificates"></td>
          </tr>

          <tr>
            <td>Address:</td>
            <td>
              <textarea name="address" rows="4"
cols="30"></textarea>
            </td>
          </tr>

          <tr>
            <td colspan="2" align="center">
              <input type="submit" value="Submit">
              <input type="reset" value="Reset">
            </td>
          </tr>
        </form>
```

```
   </table>

</body>
</html>
```

OUTPUT:



2. Aim : Create two different forms in one webpage using Iframes.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Two Forms Using Iframes</title>
</head>
<body>
  <p>Name:SUJANKR7 Rollno:227</p>

  <center>
    <h2>Two Forms Using Iframes</h2>

    <!-- Login Form Iframe -->
    <iframe width="600" height="500"
      srcdoc='
      <!DOCTYPE html>
      <html>
      <head>
        <title>User Login</title>
      </head>
      <body>
        <center>
```

```
            <h2>User Login</h2>
            <form>
              <table width="50" border="1" cellpadding="25"
cellspacing="2" style="text-align:center;">

                <tr>
                  <td>
                    <label for="username">Username:</label>
                  </td>
                  <td>
                    <input type="text" id="username"
name="username">
                  </td>
                </tr>

                <tr>
                  <td>
                    <label for="password">Password:</label>
                  </td>
                  <td>
                    <input type="password" id="password"
name="password">
                  </td>
                </tr>
```

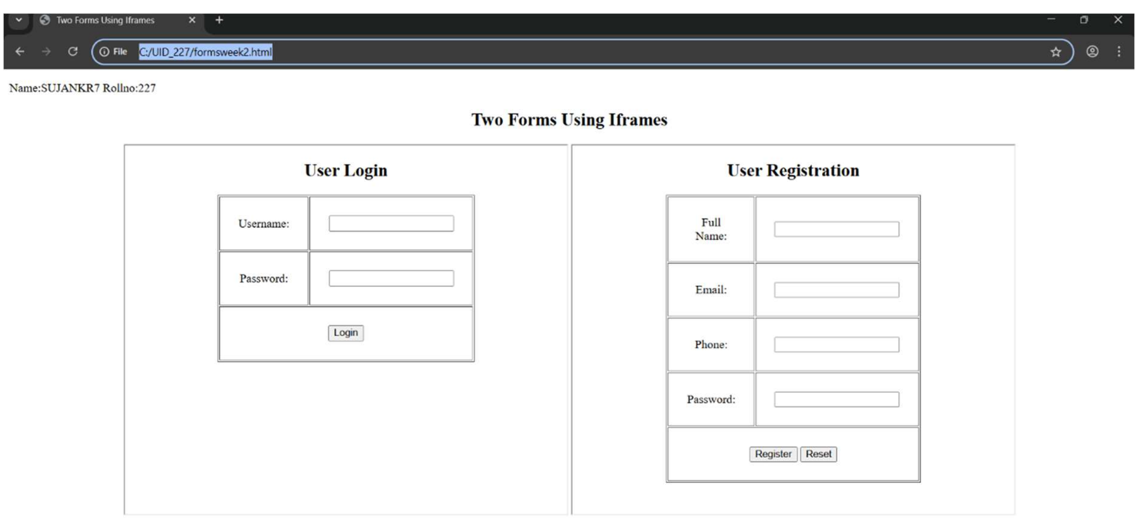
```
            <tr>
              <td colspan="2" style="text-align:center;">
                <input type="submit" value="Login">
              </td>
            </tr>

          </table>
        </form>
      </center>
    </body>
    </html>
    '>
</iframe>

<!-- Registration Form Iframe -->
<iframe width="600" height="500"
    srcdoc='
    <!DOCTYPE html>
    <html>
    <head>
      <title>User Registration</title>
    </head>
    <body>
      <center>
```

```
        <h2>User Registration</h2>
        <form>
          <table width="50" border="1" cellpadding="25"
cellspacing="2" style="text-align:center;">

            <tr>
              <td>
                <label for="fullname">Full Name:</label>
              </td>
              <td>
                <input type="text" id="fullname"
name="fullname">
              </td>
            </tr>

            <tr>
              <td>
                <label for="email">Email:</label>
              </td>
              <td>
                <input type="email" id="email" name="email">
              </td>
            </tr>
```

```
            <tr>
              <td>
                <label for="phone">Phone:</label>
              </td>
              <td>
                <input type="text" id="phone" name="phone">
              </td>
            </tr>

            <tr>
              <td>
                <label for="password">Password:</label>
              </td>
              <td>
                <input type="password" id="password"
name="password">
              </td>
            </tr>

            <tr>
              <td colspan="2" style="text-align:center;">
                <input type="submit" value="Register">
                <input type="reset" value="Reset">
              </td>
```

```
                </tr>

              </table>
            </form>
          </center>
        </body>
        </html>
        '>
      </iframe>

    </center>

</body>
</html>
```

OUTPUT:



Week-5

5. A) Aim: creating a card layout using

   tag consider three cards and a button using html and css.

CODE:

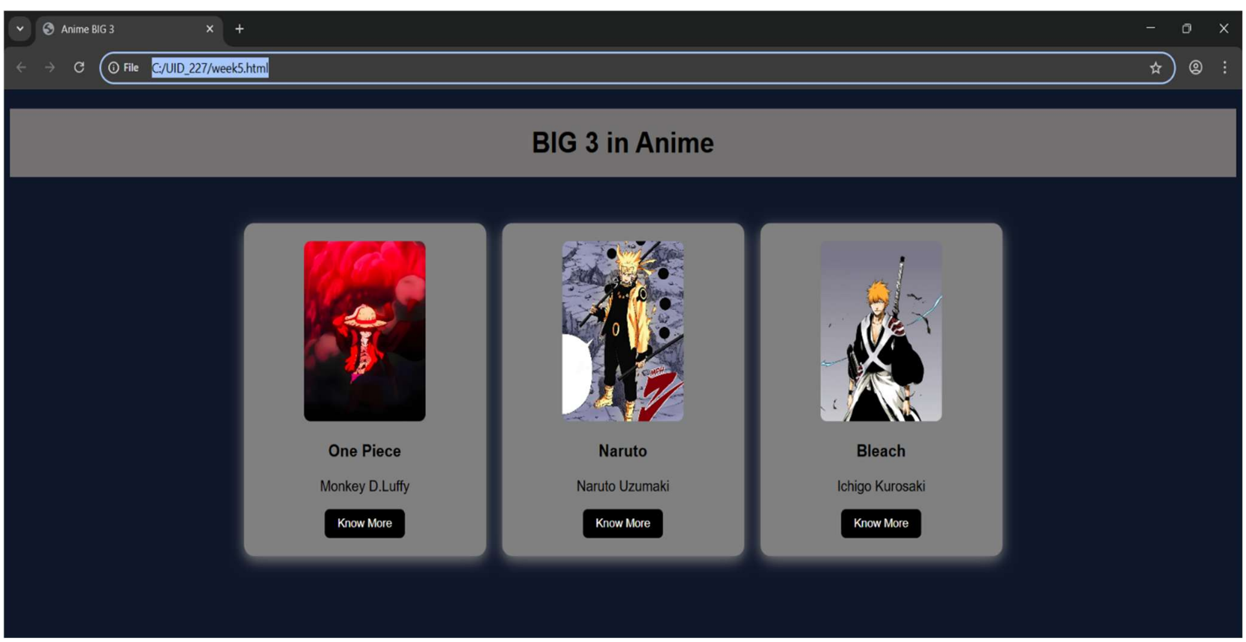
```
<!DOCTYPE html>
<html>
<head>
<title>Anime BIG 3</title>
<style>
body{
   font-family:Arial;
   background:#0f172a;
   text-align:center;
}
h1{
   background:#737171;
   padding:20px;
}
.container{
   display:flex;
   flex-wrap:wrap;
   justify-content:center;
   gap:20px;
```

```
  padding:30px;
}
.card{
  background: gray;
  color:black;
  width:260px;
  padding:20px;
  border-radius:12px;
  box-shadow:0 5px 15px gray;
}
.hero-img{
  width:150px;
  height:200px;
  object-fit:cover;
  border-radius:8px;
}
button{

  padding:8px 16px;
  border:none;
  background:black;
```

```
   color:white;
   border-radius:6px;
}
button:hover{
   background:black;
}
</style>
</head>
<body>
<h1>BIG 3 in Anime</h1>
<div class="container">
   <div class="card">
      <img class="hero-img"
       src="c:\Users\sujan\Downloads\Luffy 4k Wallpaper One
Piece.jpg">
      <h3>One Piece</h3>
      <p>Monkey D.Luffy</p>
      <button>Know More</button>
   </div>
   <div class="card">
      <img class="hero-img"
       src="c:\Users\sujan\Downloads\download (3).jpg">
      <h3>Naruto</h3>
      <p>Naruto Uzumaki</p>
```

```
      <button>Know More</button>
    </div>
    <div class="card">
      <img class="hero-img"
       src="c:\Users\sujan\Downloads\Kurosaki Ichigo.jpg">
      <h3>Bleach</h3>
      <p>Ichigo Kurosaki</p>
      <button>Know More</button>
    </div>
  </div>
  </body>
  </html>
```

OutPut:



Explanation :

<!DOCTYPE html> : This declaration tells the browser that the document is an HTML5 document and should be rendered using the latest HTML standard.

<html> : This is the root element of the webpage. All other HTML elements are placed inside this tag.

<head> : The <head> section contains metadata about the webpage such as the title and CSS styling. It does not display content directly on the page.

<title> : The <title> tag defines the name of the webpage that appears on the browser tab.

<body> : The <body> tag contains all the visible content of the webpage such as text, images, buttons, and other elements.

<h1> / <h2> / <h3> : Heading tags are used to display titles and subtitles. They help structure the content of the webpage.

<p> : The paragraph tag is used to display blocks of text.

<div> : The <div> element is used to group and organize content into sections. It helps in structuring and styling different parts of the webpage.

<button> : The <button> tag creates a clickable button on the webpage.

<img> : The <img> tag is used to display images. The src attribute specifies the path or location of the image file.

We used internal CSS inside the <style> tag within the <head> section.
Element selectors and class selectors were used to apply styles to specific elements.By using classes and CSS styling, we designed and structured the webpage layout effectively.

5. B) Aim: Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page.

<u>CODE:</u>

[Index.html]

```
<!DOCTYPE html>
<html>
<head>
<title>Luffy - Gear Navigation</title>

<style>
body{
   margin:0;
   font-family:Arial;
   background-color: lightgray;
}

.menu{
   background:#e11d48;
   padding:15px;
   text-align:center;
}
```

```
.menu a{
   color:white;
   margin:0 15px;
   text-decoration:none;
   background:black;
   padding:8px 12px;
}

iframe{
   width:100%;
   height:400px;
   border:none;
}
</style>

</head>

<body>

<h1 style="text-
align:center;background:#e11d48;color:white;padding:20px;">
Monkey D. Luffy - 4 Gears
</h1>
```

```
<div class="menu">
   <a href="gear2.html" target="frame">Gear 2</a>
   <a href="gear3.html" target="frame">Gear 3</a>
   <a href="gear4.html" target="frame">Gear 4</a>
   <a href="gear5.html" target="frame">Gear 5</a>
</div>

<iframe name="frame"></iframe>


</body>
</html>
[gear_2]
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
   border:2px solid black;
   padding:20px;
   display:inline-block;
}
h2{color:#ef4444;}
</style>
```

```
</head>

<body>
<div class="box">
<h2>Gear 2</h2>
<p>Boosts speed and strength by increasing blood flow.</p>
<img src="luffy.jpg" width="200">
</div>
</body>
</html>
[gear_3]
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
   border:2px solid black;
   padding:20px;
   display:inline-block;
}
h2{color:#3b82f6;}
</style>
</head>
```

```
<body>
<div class="box">
<h2>Gear 3</h2>
<p>Inflates bones to create giant powerful attacks.</p>
<img src="gear3.jpg.jpg" width="200">
</div>
</body>
</html>
```

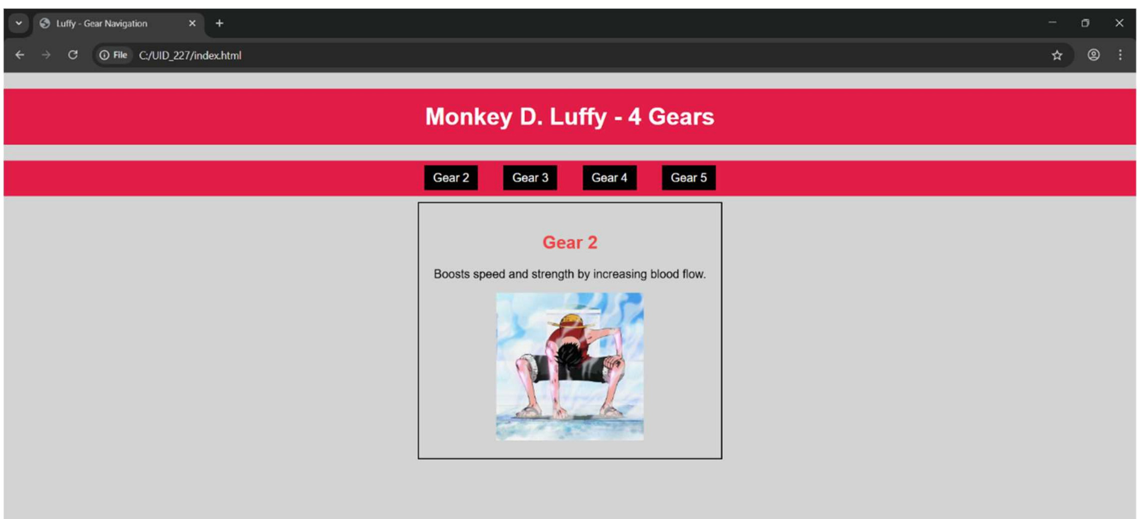
[gear_4]

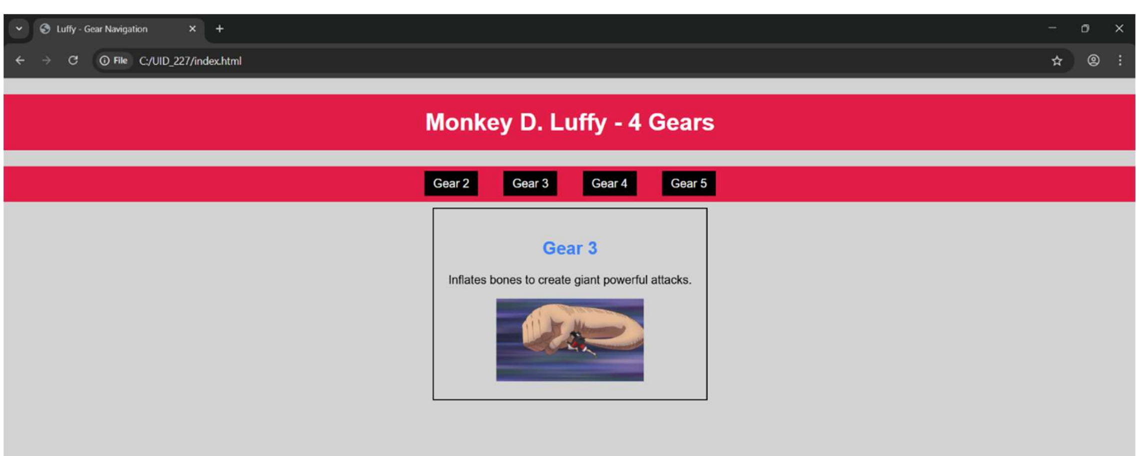
```
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
   border:2px solid black;
   padding:20px;
   display:inline-block;
}
h2{color:#7c3aed;}
</style>
</head>
```

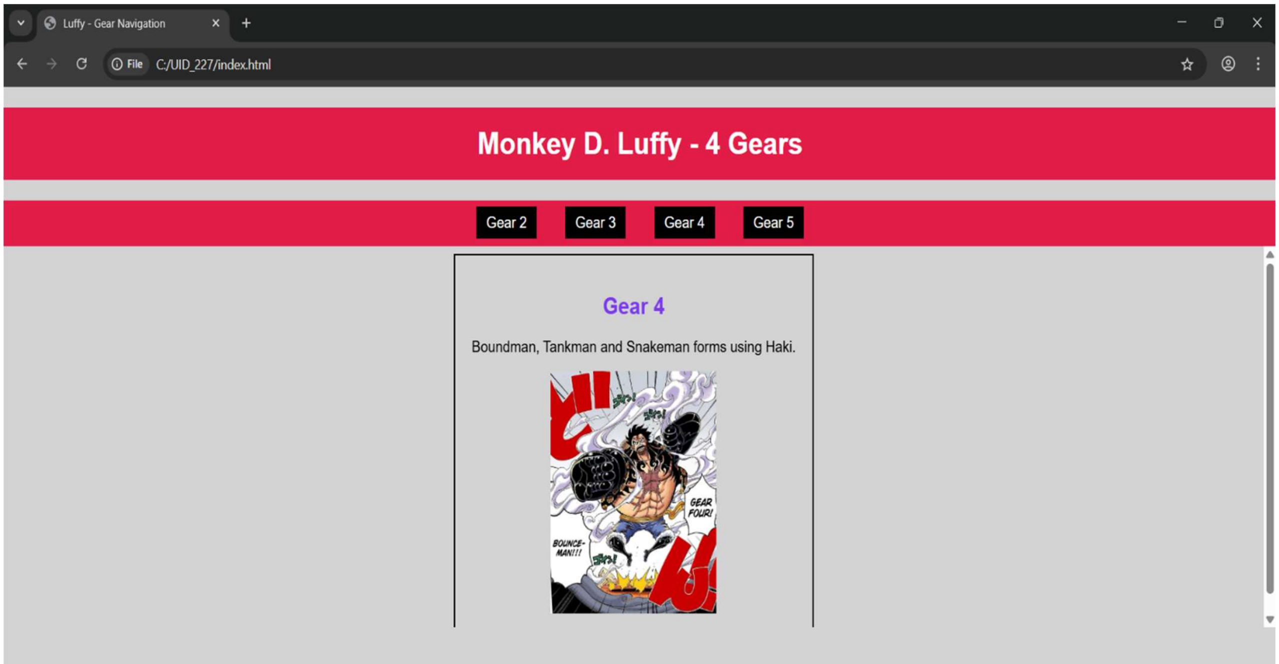
```
<body>
<div class="box">
<h2>Gear 4</h2>
<p>Boundman, Tankman and Snakeman forms using Haki.</p>
<img src="download (4).jpg" width="200">
</div>
</body>
</html>
[gear_5]
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
   border:2px solid black;
   padding:20px;
   display:inline-block;
}
h2{color:#9ca3af;}
</style>
</head>

<body>
```

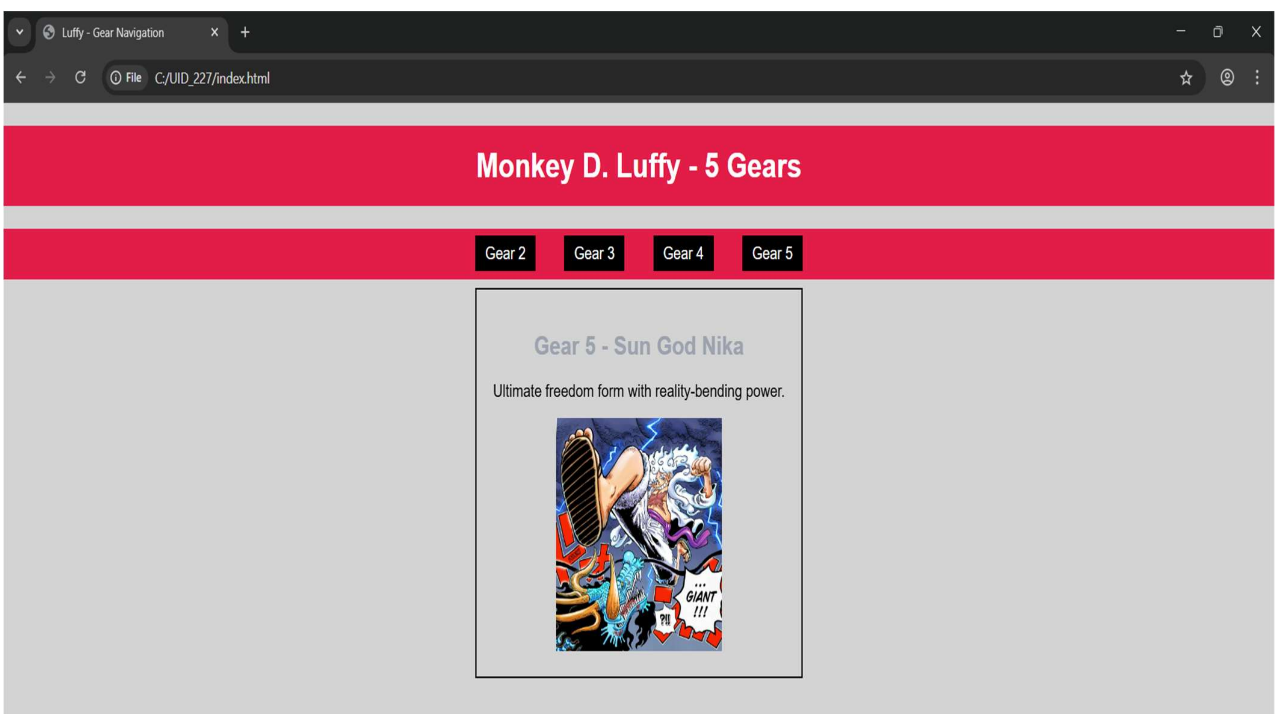
```
<div class="box">
<h2>Gear 5 - Sun God Nika</h2>
<p>Ultimate freedom form with reality-bending power.</p>
<img src="gear5.jpg" width="200">
</div>
</body>
</html>
```

<u>OutPut:</u>






Luffy - Gear Navigation
File C:/UID_227/index.html
Monkey D. Luffy - 4 Gears
Gear 2
Gear 3
Gear 4
Gear 5
Gear 4
Boundman, Tankman and Snakeman forms using Haki.
GEAR FOUR!
BOUNCE-MAN!!!


Luffy - Gear Navigation
File C:/UID_227/index.html
Monkey D. Luffy - 5 Gears
Gear 2
Gear 3
Gear 4
Gear 5
Gear 5 - Sun God Nika
Ultimate freedom form with reality-bending power.
... GIANT !!!
?!!

Explanation :

<!DOCTYPE html> : The <!DOCTYPE html> declaration tells the browser that the document is an HTML5 document and helps it display the webpage correctly.

<html> : The <html> element is the root element of the webpage. All other elements are placed inside this tag.

<head> : The <head> section contains metadata about the webpage such as title and styling information.

<title> : The <title> tag defines the title of the webpage, which appears in the browser tab.

<body> : The <body> element contains all the visible content of the webpage like text, images, links, tables, etc.

<h1> / <h2> : The heading tags are used to display headings on the webpage.

<p> : The <p> tag is used to write paragraphs.

<div> : The <div> element is used to group content into sections so that it can be styled or organized easily.

<a> : The <a> (anchor) tag is used to create links between webpages.

<iframe> : The <iframe> element is used to display another HTML page within the main webpage.

<img> : The <img> tag is used to display images on the webpage using the src attribute.

We used internal CSS inside the <style> tag in the head section to design the webpage. Element selectors and class selectors are used to apply styles to different sections. The href attribute in the anchor tag links the other HTML pages to the main page, and the target attribute is used to display those pages inside the iframe. The border styling is applied using inline CSS to create a visible box around the content.

By combining HTML structure and CSS styling, we created a mini website where multiple pages are connected to the main webpage using iframe navigation.

5. c) Aim: Design amrita login page by showcampus on home page.upon clicking on each campus the details about each campus should be display on the same page use html and css to design a stylish webpage.

CODE:

[Amrita]

```
<!DOCTYPE html>

<html>

Name=sujan rollno:227

<head>

<title>Amrita Vishwa Vidhyapeetam</title>

<style>

body{margin:0;font-family:Arial;background:rgb(101, 101, 139);}

header{background:#0a0202;color:#fff;text-align:center;

   padding:20px;font-size:36px;}

nav{background:#f7f2f5;text-align:center;padding:10px;}

nav a{color:rgb(231, 23, 169);margin:12px;text-decoration:none;

   font-weight:bold;padding:5px;}

nav a:hover{background:#fff;color:#0b5ea6;

   border-radius:5px;}

main{display:flex;padding:20px;}

.login{
```

```
    width:350px;
    background:#787676;
    padding:15px;
    border-radius:10px;
    color:rgb(19, 18, 18);
}
.login input[type="email"],
.login input[type="password"]{
    width:90%;
    padding:6px;
    margin:6px 0;
}
.remember{display:flex;align-items:center;gap:6px;margin:8px 0;}
iframe{
    flex:1;
    margin-left:20px;
    height:520px;
    border:none;
    background:lightskyblue;
    border-radius:10px;
}
</style>
</head>
```

```
<body>
<header>AMRITA VISHWA VIDHYAPEETAM</header>
<nav>
<a href="amaravati.html" target="campus">AMARAVATI</a>
<a href="amritapuri.html" target="campus">AMRITAPURI</a>
<a href="bengaluru.html" target="campus">BENGALURU</a>
<a href="chennai.html" target="campus">CHENNAI</a>
<a href="coimbatore.html"
target="campus">COIMBATORE</a>
<a href="faridabad.html" target="campus">FARIDABAD</a>
<a href="kochi.html" target="campus">KOCHI</a>
<a href="mysuru.html" target="campus">MYSURU</a>
</nav>
<main>
<div class="login">
<h2>Login</h2>
Email
<input type="email">
Password
<input type="password">
<div class="remember">
<input type="checkbox"><label>Save me</label>
</div>
```

```
<input type="button" value="Login"
style="width:100%;padding:6px;">

<br><br>

<a href="#">Forgot Password?</a><br>

No account? <a href="#">Register</a>

</div>

<iframe name="campus"
src="c:\Users\sujan\Downloads\amrita.amaravathi.jpg"></ifra
me>

</main>

</body>

</html>
```

[amaravathi]

```
<!DOCTYPE html>

<html>

<body style="font-family:Arial; text-align:center;
padding:30px;">

<h1 style="color:#a00000;">Amaravati Campus</h1>

<p>

Amrita Amaravati campus is located near the Krishna River in
Andhra Pradesh.

It provides modern classrooms, research labs, and excellent
hostel facilities.

</p>

<p>
```

```
Popular Courses: CSE, AI, Data Science<br>
Facilities: Library, Smart Classrooms, Hostel<br>
Focus: Innovation and Research
</p>
<img src="c:\Users\sujan\Downloads\amrita.amaravathi.jpg"
width="450">
</body>
</html>
```

[Bengaluru]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-
align:center;padding:30px;">
<h1 style="color:#a00000;">Bengaluru Campus</h1>
<p>
Located in India's tech hub, Bengaluru campus offers
advanced
engineering programs with strong industry connections.
</p>
<p>
Popular Courses: Robotics, CSE, ECE<br>
Facilities: Coding Labs, Innovation Centres<br>
Focus: Industry Collaboration
</p>
```

```
<img src="c:\Users\sujan\Downloads\amrita.blr.jpg"
width="450">
</body>
</html>
```

[Amrithapuri]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Amritapuri Campus</h1>
<p>
Amritapuri campus in Kerala is the main campus of Amrita University,
located near the Arabian Sea with a peaceful academic atmosphere.
</p>
<p>
Popular Courses: Engineering, Arts, Sciences<br>
Facilities: Research Labs, Beach Campus, Hostel<br>
Focus: Academic Excellence with Values
</p>
<img src="c:\Users\sujan\Downloads\amrithapuri.jpg"
width="450">
</body>
</html>
```

[Chennai]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-
align:center;padding:30px;">
<h1 style="color:#a00000;">Chennai Campus</h1>
<p>
Chennai campus provides top-quality computing and
management education
with modern infrastructure and placement support.
</p>
<p>
Popular Courses: IT, MBA, Electronics<br>
Facilities: Digital Library, Placement Cell<br>
Focus: Career Development
</p>
<img src="c:\Users\sujan\Downloads\amrita.chennai.jpg"
width="450">
</body>
</html>
```

[Coiambatore]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-
align:center;padding:30px;">
```

```
<h1 style="color:#a00000;">Coimbatore Campus</h1>
<p>
Coimbatore campus is well known for its engineering excellence
and research-oriented education.
</p>
<p>
Popular Courses: Mechanical, Civil, CSE<br>
Facilities: Workshops, Labs, Hostel<br>
Focus: Core Engineering Skills
</p>
<img src="c:\Users\sujan\Downloads\amrita.chennai.jpg"
width="450">
</body>
</html>
```

[Fairabadh]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Faridabad Campus</h1>
<p>
Faridabad campus focuses on medical and healthcare education
with advanced hospital training facilities.
```

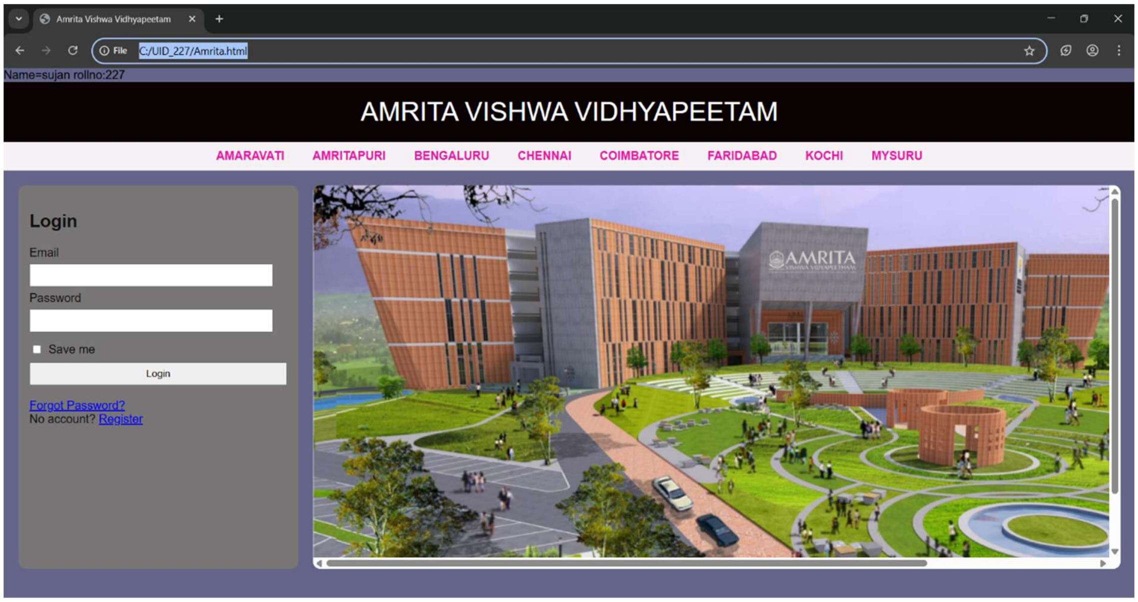
```
</p>
<p>
Popular Courses: Medical, Nursing, Allied Health<br>
Facilities: Teaching Hospital, Labs<br>
Focus: Healthcare Research
</p>
<img src="c:\Users\sujan\Downloads\amrita.faridabad.jpg"
width="450">
</body>
</html>
[Kochi]
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-
align:center;padding:30px;">
<h1 style="color:#a00000;">Kochi Campus</h1>
<p>
Kochi campus offers programs in business, commerce,
and applied sciences with modern facilities.
</p>
<p>
Popular Courses: MBA, Commerce, Finance<br>
Facilities: Seminar Halls, Library<br>
Focus: Industry Training
```

```
</p>
<img src="c:\Users\sujan\Downloads\amrita.kochi.jpg" width="450">
</body>
</html>
```

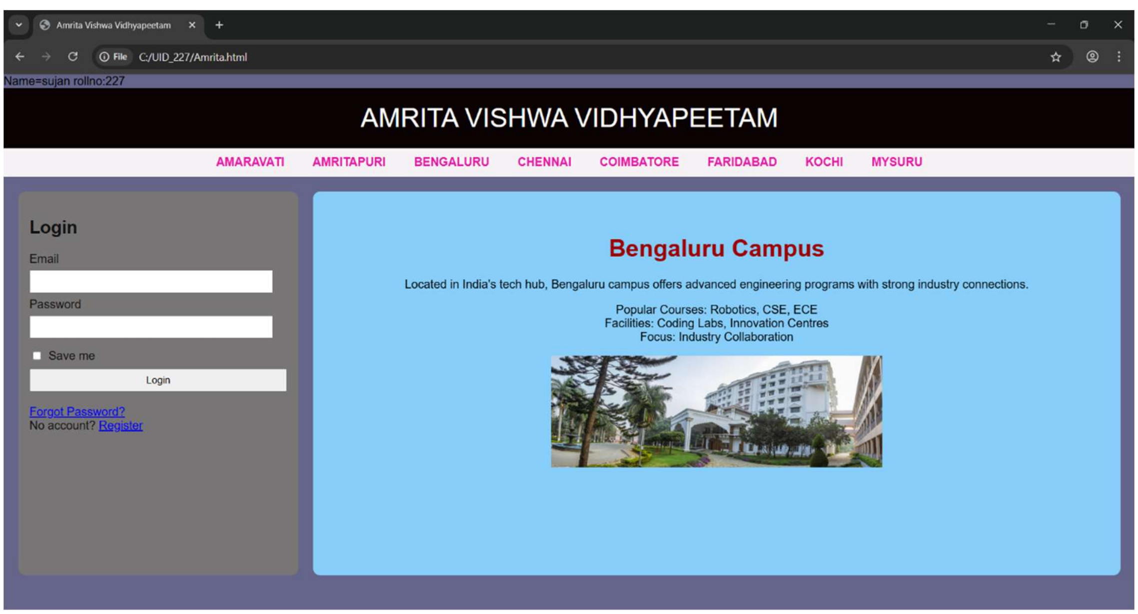
[Mysore]

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Mysuru Campus</h1>
<p>
Mysuru campus is known for its peaceful environment,
green campus, and quality education.
</p>
<p>
Popular Courses: Arts, Sciences, Engineering<br>
Facilities: Hostel, Library, Sports Complex<br>
Focus: Calm Learning Atmosphere
</p>
<img src="c:\Users\sujan\Downloads\amrita.mysore.jpg" width="450">
</body>
</html>
```

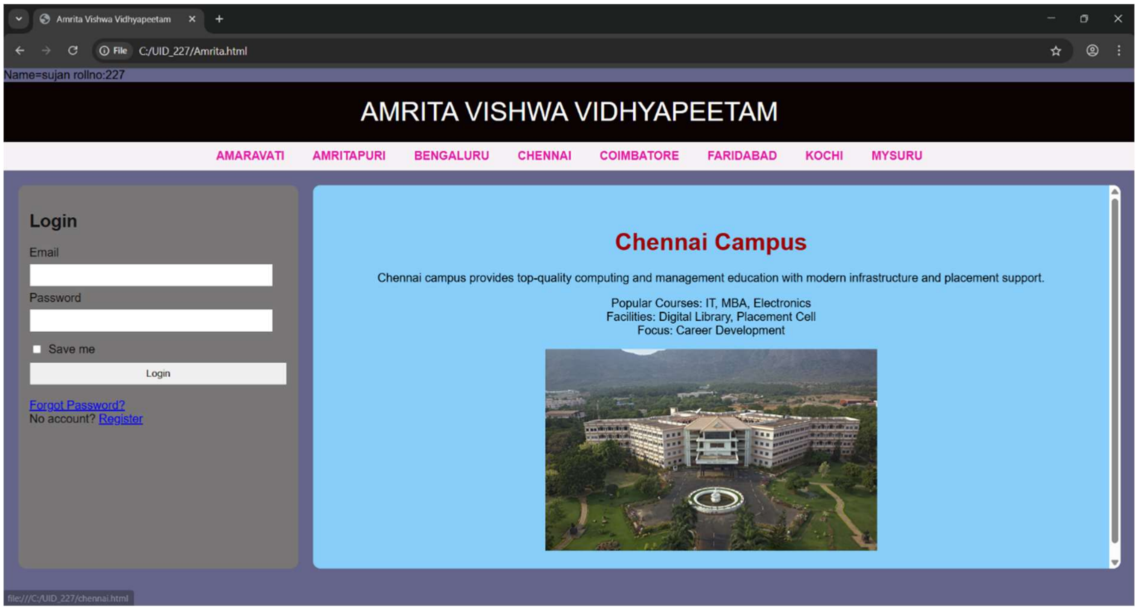
OutPut;

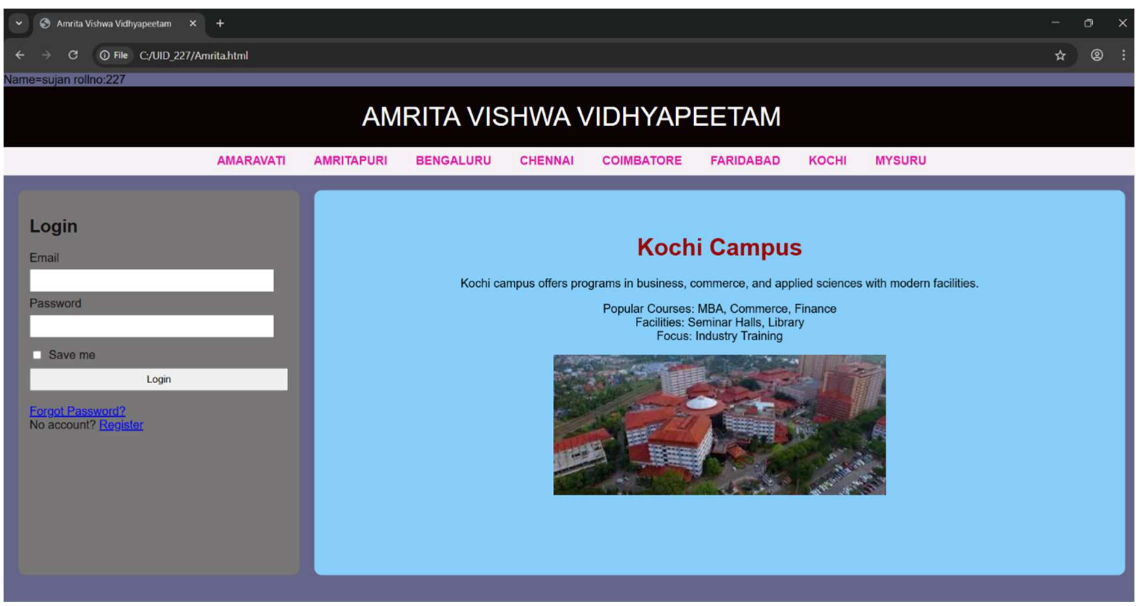


Amrita Vishwa Vidhyapeetam
C:/UID_227/Amrita.html
Name=sujan rollno:227
AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
AMRITA



Amrita Vishwa Vidhyapeetam
C:/UID_227/Amrita.html
Name=sujan rollno:227
AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Bengaluru Campus
Located in India's tech hub, Bengaluru campus offers advanced engineering programs with strong industry connections.
Popular Courses: Robotics, CSE, ECE
Facilities: Coding Labs, Innovation Centres
Focus: Industry Collaboration


Amrita Vishwa Vidhyapeetam
C:/UID_227/Amrita.html
Name=sujan rollno:227
AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Chennai Campus
Chennai campus provides top-quality computing and management education with modern infrastructure and placement support.
Popular Courses: IT, MBA, Electronics
Facilities: Digital Library, Placement Cell
Focus: Career Development


Amrita Vishwa Vidhyapeetam
C:/UID_227/Amrita.html
Name=sujan rollno:227
AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Kochi Campus
Kochi campus offers programs in business, commerce, and applied sciences with modern facilities.
Popular Courses: MBA, Commerce, Finance
Facilities: Seminar Halls, Library
Focus: Industry Training

Week-6

6) Aim: Design a responsive e-commerc application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

CODE;

Main:

Name: Sujan

Roll No: 25227

Section: C

```
<!DOCTYPE html>
<html>
<head>
<title>My Ecommerce Store</title>

<style>

body{
font-family: Arial;
margin:0;
background:#f5f5f7;
}
```

```
.header{
background:#0f172a;
color:white;
text-align:center;
padding:20px;
font-size:28px;
}

.sale{
background:#2563eb;
color:white;
text-align:center;
padding:10px;
font-size:18px;
font-weight:bold;
}

.container{
display:flex;
min-height:80vh;
}

.sidebar{
width:200px;
```

```
background:#1e293b;
color:white;
padding:20px;
}

.sidebar h3{
margin-top:0;
}

.sidebar a{
display:block;
color:white;
text-decoration:none;
padding:10px 0;
border-bottom:1px solid #334155;
}

.sidebar a:hover{
color:#60a5fa;
}

.frame{
flex:1;
border:none;
```

```
}

.footer{
background:#0f172a;
color:white;
text-align:center;
padding:15px;
}

</style>
</head>

<body>

<div class="header">
My Ecommerce Store
</div>

<div class="sale">
Big Sale! Up to 50% Off
</div>

<div class="container">
```

```
<div class="sidebar">
<h3>Categories</h3>

<a href="ecomelec.html" target="frame">Electronics</a>
<a href="ecomcloth.html" target="frame">Clothes</a>
<a href="ecomacces.html" target="frame">Accessories</a>

</div>

<iframe class="frame" name="frame"
src="electronics.html"></iframe>

</div>

<div class="footer">
© 2026 My Ecommerce Store
</div>

</body>
</html>
```

Electronics

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>Electronics</title>
<style>
  body {
    margin: 0;
    background-color: aliceblue;
    font-family: sans-serif;
  }


  .products {
    grid-area: products;
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 10px;
    padding: 20px;
  }

  .product-card {
    background: white;
    border: 1px solid #ddd;
    padding: 20px;
    text-align: center;
    border-radius: 8px;
    height: auto;
```

```
    width: 600px;
}
.p{
    text-align: center;
    background-color: tomato;
    align-items: center;
}
img{
    width: 120px;
    height: auto;
    border-radius: 8px;
}


.button {
    background-color: blue;
    color: white;
    border: none;
    padding: 10px 20px;
    cursor: pointer;
    border-radius: 50px;
    margin-top: 10px;
}
```

```
    .button:hover {
      background-color: #7094ff;
    }
  </style>
</head>
<body>
    <div class="products">
      <div class="product-card">
        <img src="c:\Users\sujan\OneDrive\Pictures\iphone.jpg">
        <h3>Iphone 17 pro max</h3>
        <p>$899</p>
        <p>12GB RAM | 1TB Storage</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="c:\Users\sujan\OneDrive\Pictures\sams26.jpg">
        <h3>Samsung Galaxy S26</h3>
        <p>$1199</p>
        <p>12GB RAM | 512GB Storage</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product-card">
        <img src="c:\Users\sujan\OneDrive\Pictures\nord5.avif">
```

```
        <h3>One PLus Nord 5</h3>
        <p>$299</p>
        <p>12GB RAM | 256GB Storage</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="c:\Users\sujan\OneDrive\Pictures\nothing3a.jpg">
        <h3>Nothing 3a</h3>
        <p>$249</p>
        <p>12GB RAM | 256GB Storage</p>
        <button class="button">Add to Cart</button>
      </div>
    </div>
</body>
```

Clothes:

```
<!DOCTYPE html>
<html>
<head>
  <title>Electronics</title>
  <style>
    body {
      margin: 0;
```

```css
  background-color: aliceblue;
  font-family: sans-serif;
}


.products {
  grid-area: products;
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 20px;
  padding: 20px;
}

.product-card {
  background: white;
  border: 1px solid #ddd;
  padding: 20px;
  text-align: center;
  border-radius: 8px;
}
img{
  width: 120px;
  height: auto;
  border-radius: 8px;
```

```
        }
        .p{
          text-align: center;
          background-color: tomato;
          align-items: center;
        }


        .button {
          background-color: blue;
          color: white;
          border: none;
          padding: 10px 20px;
          cursor: pointer;
          border-radius: 50px;
          margin-top: 10px;
        }

        .button:hover {
          background-color: #7094ff;
        }
      </style>
    </head>
    <body>
```

```
    <div class="products">
      <div class="product-card">
        <img
src="c:\Users\sujan\OneDrive\Pictures\blueshirt.avif">
        <h3>H&M blue shirt</h3>
        <p>$39</p>
        <p>Size: S , M , XL , XXL</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="c:\Users\sujan\OneDrive\Pictures\blackshirt.jpg">
        <h3>H&M Black shirt</h3>
        <p>$59</p>
        <p>Size: S , M , L , XL</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product-card">
        <img src="c:\Users\sujan\OneDrive\Pictures\black
jean.avif">
        <h3>H&M black pants</h3>
        <p>$49</p>
        <p>Size: 28 , 30 , 32 , 34</p>
        <button class="button">Add to Cart</button>
      </div>
```

```
      <div class="product-card">
        <img src="c:\Users\sujan\OneDrive\Pictures\blue
jean.jpg">
        <h3>USPA blue pants</h3>
        <p>$52</p>
        <p>Size: 28 , 30 , 32 , 34</p>
        <button class="button">Add to Cart</button>
      </div>
    </div>
</body>
```

Accescories:

```
<!DOCTYPE html>
<html>
<head>
  <title>Electronics</title>
  <style>
    body {
      margin: 0;
      background-color: aliceblue;
      font-family: sans-serif;
    }
```

```
.products {
  grid-area: products;
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 20px;
  padding: 20px;
}

.product-card {
  background: white;
  border: 1px solid #ddd;
  padding: 20px;
  text-align: center;
  border-radius: 8px;
}
.p{
  text-align: center;
  background-color: tomato;
  align-items: center;
}
img{
  width: 120px;
  height: auto;
```

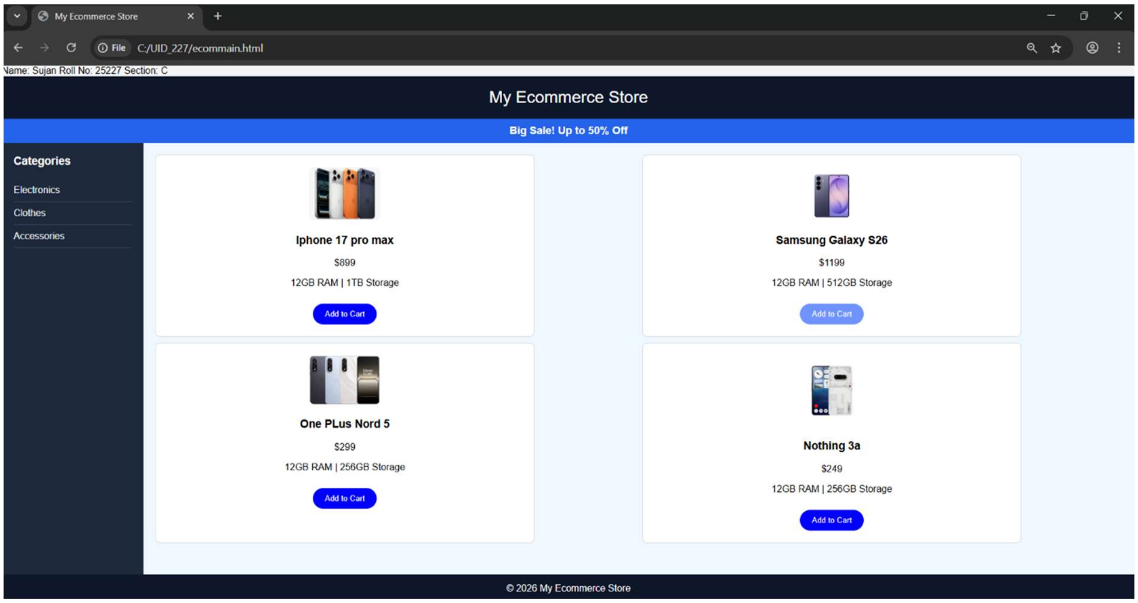
```
        border-radius: 8px;
    }


    .button {
        background-color: blue;
        color: white;
        border: none;
        padding: 10px 20px;
        cursor: pointer;
        border-radius: 50px;
        margin-top: 10px;
    }

    .button:hover {
        background-color: #7094ff;
    }
  </style>
</head>
<body>
    <div class="products">
        <div class="product-card">
            <img src="c:\Users\sujan\OneDrive\Pictures\Samsung
Galaxy Watch 6.jpg">
```

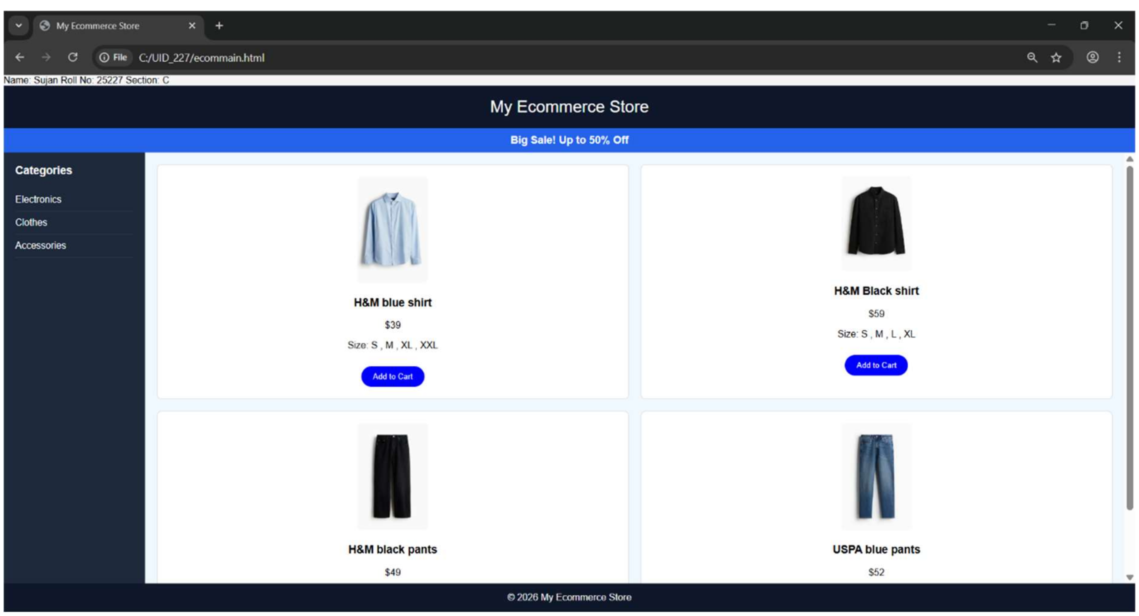
```
          <h3>Samsung Galaxy Watch 6</h3>
          <p>$169</p>
          <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
          <img src="c:\Users\sujan\OneDrive\Pictures\Samsung
Galaxy Buds 4.jpg" >
          <h3>Samsung Galaxy Buds 4</h3>
          <p>$249</p>
          <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
          <img src="c:\Users\sujan\OneDrive\Pictures\Samsung
Galaxy ring.avif">
          <h3>Samsung Galaxy ring </h3>
          <p>$459</p>
          <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
          <img src="c:\Users\sujan\OneDrive\Pictures\i
pocket.jpg">
          <h3>i pocket </h3>
          <p>$59</p>
          <button class="button">Add to Cart</button>
        </div>
```

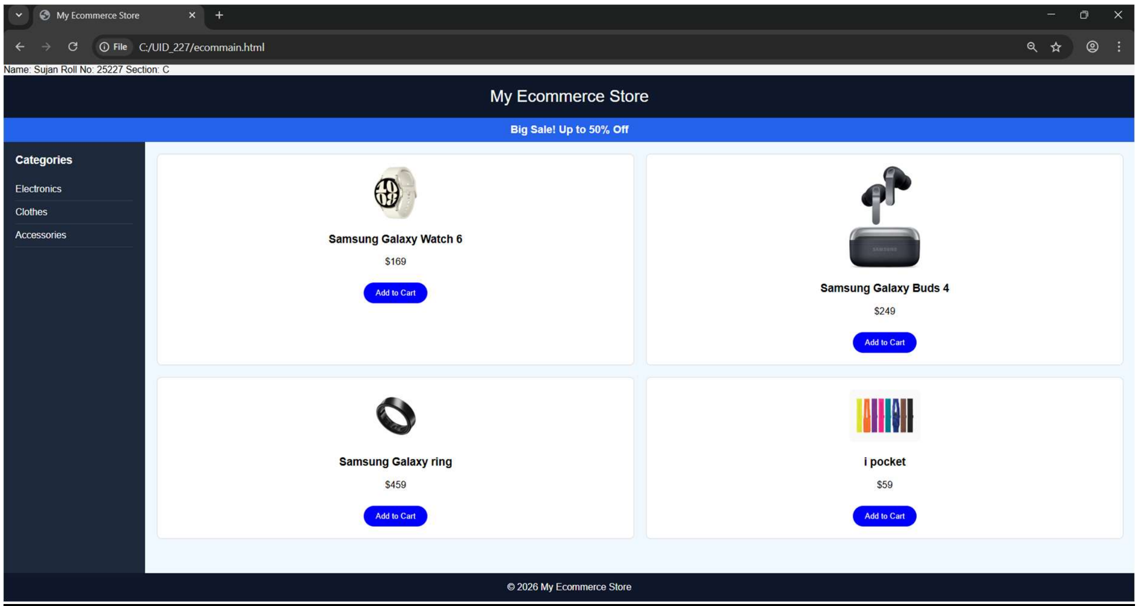
</div>

</body>

Output:






My Ecommerce Store
C:/UID_227/ecommain.html
Name: Sujan Roll No: 25227 Section: C
My Ecommerce Store
Big Sale! Up to 50% Off
Categories
Electronics
Clothes
Accessories
Samsung Galaxy Watch 6
$169
Add to Cart
Samsung Galaxy Buds 4
$249
Add to Cart
Samsung Galaxy ring
$459
Add to Cart
i pocket
$59
Add to Cart
© 2026 My Ecommerce Store

Week-7

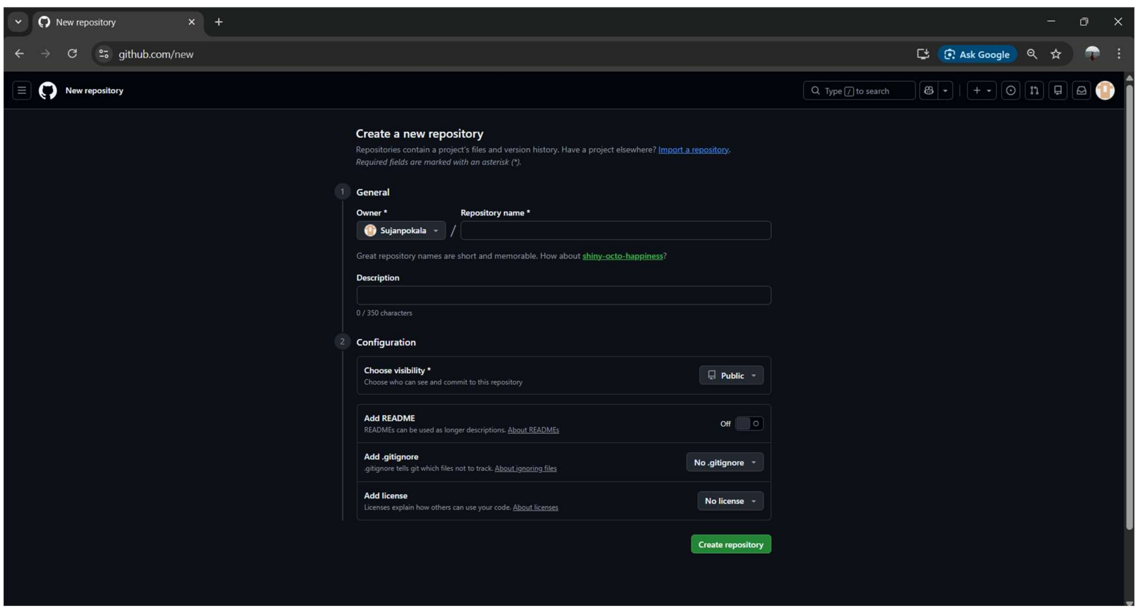
7. Aim: create a GitHub account and create a repository in it and push all the week 6 code files in it.

Procedure :

Step 1: browse for GitHub and enter it.

Step 2:login to GitHub account.

Step 3:create a repository after login by clicking new.



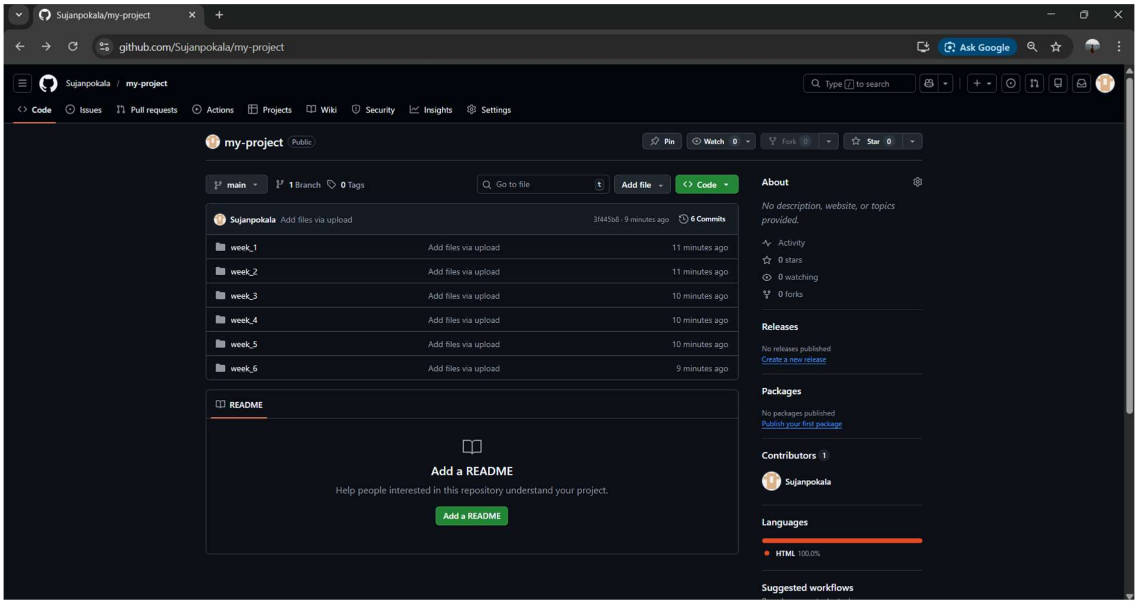
Step 4:specify the repository specifications.

Step 5:download GitHub desktop and Lauch it .

Step 6:add the repository to it .

Step 7:add the vs code file to the GitHub desktop.

Step 8:add commit to it and the files to repository and sync the changes . it automatically pushes the files .



Step 9:now go to browser and check it . it will show all the new added files in the repository .

My repository link for the uid all week's work:

https://github.com/Sujanpokala/my-project

Week-8

8.A) Aim: create a web page that displays a square that continuously rotates 360 using css animation.

Program :

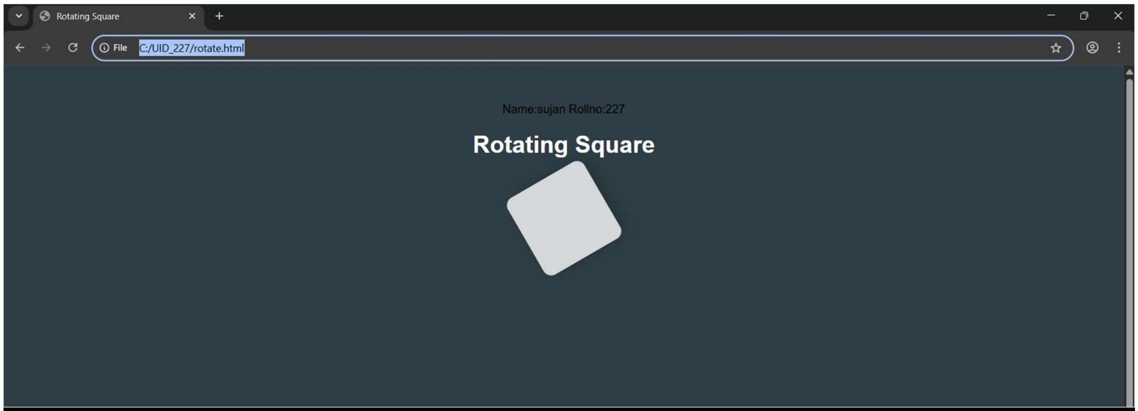
Name:sujan

Rollno:227

```
<!DOCTYPE html>
<html>
<head>
 <title>Rotating Square</title>
 <style>
  body {
   margin: 0;
   height: 100vh;
   display: flex;
   flex-direction: column;
   justify-content: flex-start;
   align-items: center;
   background-color: #2f3e46;
   font-family: Arial, sans-serif;
   padding-top: 50px;
  }
  .square {
   width: 120px;
```

```
    height: 120px;
    background: #ffffffcc;
    border-radius: 12px;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.2);
    animation: rotate 3s linear infinite;
   }
   @keyframes rotate {
    from {
     transform: rotate(0deg);
    }
    to {
     transform: rotate(360deg);
    }
   }
  </style>
</head>
<body>
   <h1 style="color: #fff;">Rotating Square</h1>
   <div class="square"></div>
</body>
</html>
```

<u>Output:</u>



<!DOCTYPE html> :
This declaration tells the browser that the document is written in HTML5 and should be displayed accordingly.

<html> :
This is the root element of an HTML page. All other elements are placed inside it.

<head> :
This section contains metadata about the webpage, such as title, styles, and other information not directly visible on the page.

<title> :
Defines the title of the webpage, which appears on the browser tab.

<body> :
Contains all the visible content of the webpage like text, images, links, tables, etc.

<h1> to <h6> :
Heading tags used to define headings, where <h1> is the largest and <h6> is the smallest.

<div> :
A container element used to group sections of content, making it easier to style and manage layouts.

@keyframes :
Used in CSS to define animations by specifying the changes in styles at different stages (from start to end).

Animation properties :
The animation shorthand property is used to apply animations by combining duration, timing, delay, and other settings in one line.

CSS usage :
Internal CSS is used within the <style> tag to design the webpage, and style attributes are applied to enhance appearance.

8.B) Aim :create a ball that moves up and down continuously using css animation.

Program :

Name:sujan

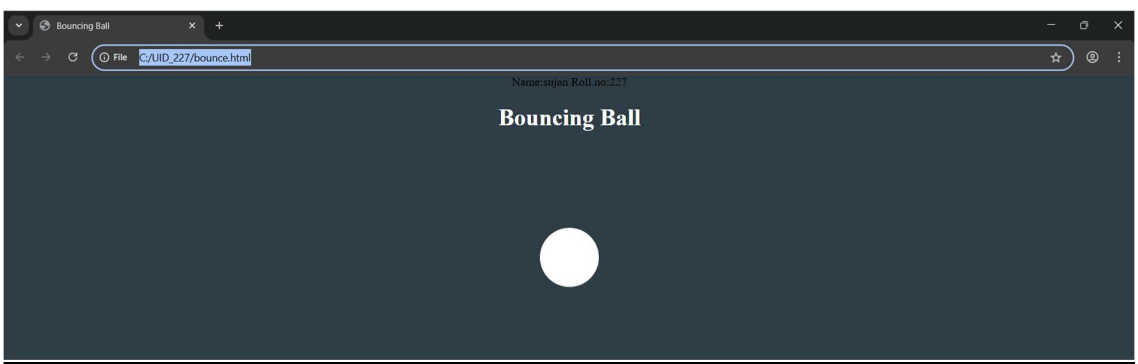
Roll.no:227

```
<!DOCTYPE html>
<html>
<head>
  <title>Bouncing Ball</title>
  <style>
    body {
      margin: 0;
      height: 100vh;
```

```
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: flex-start;
  background: #2f3e46;
}
h1 {
  margin-top: 20px;
  color: #ffffff;
}
.ball {
  width: 80px;
  height: 80px;
  background: #ffffff;
  border-radius: 50%;
  margin-top: 120px;
  animation: bounce 1.5s ease-in-out infinite;
}
@keyframes bounce {
  0% {
    transform: translateY(0);
  }
  50% {
    transform: translateY(-120px);
```

```
    }
    100% {
      transform: translateY(0);
    }
  }
 </style>
</head>
<body>
 <h1>Bouncing Ball</h1>
 <div class="ball"></div>
</body>
</html>
```

OutPut:



Explanation:

<DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML document

or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

<title> : the <title> element specifies a title for the HTML Page

<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

<div>:we use <div> element for encapsulating the the divisions and easy to modify and

@keyframes:we use @keyframes for animation from starting to end transformation.

We used animation shorthand property for these things to mention the animations preferences and

use keyframes for mentioning the positions of the animation and used style attribute and some value

properties for look and we used internal css in it .

8. .

Program:

Name:sujan

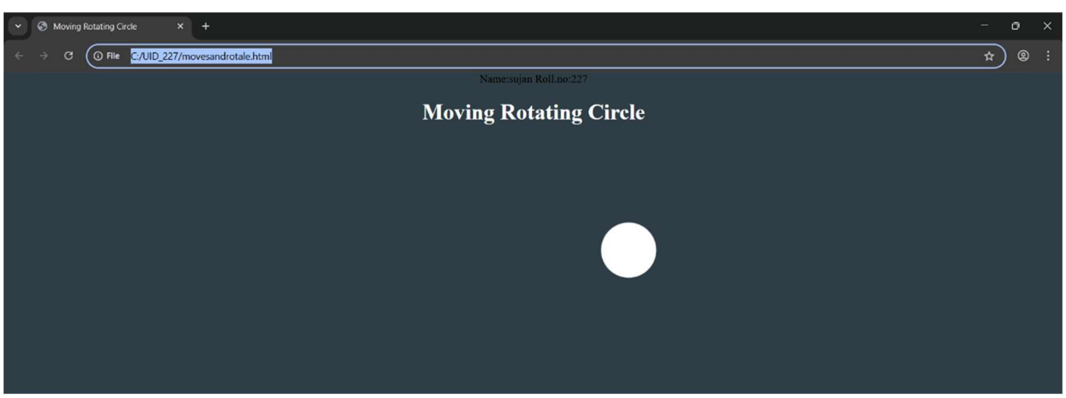
Roll.no:227

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Moving Rotating Circle</title>
```

```
<style>
 body {
  margin: 0;
  height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: flex-start;
  background: #2f3e46;
  overflow: hidden;
 }
 h1 {
  margin-top: 20px;
  color: #ffffff;
 }
 .circle {
  width: 80px;
  height: 80px;
  background: #ffffff;
  border-radius: 50%;
  margin-top: 120px;
  animation: moveRotate 3s linear infinite;
 }
 @keyframes moveRotate {
```

```
      0% {
        transform: translateX(-150px) rotate(0deg);
      }
      50% {
        transform: translateX(150px) rotate(180deg);
      }
      100% {
        transform: translateX(-150px) rotate(360deg);
      }
    }
  </style>
</head>
<body>
  <h1>Moving Rotating Circle</h1>


  <div class="circle"></div>
</body>
</html>
```

OutPut:



Week-9

9. Aim: Week 9:Create a webpage using html,css,js based on the give conditions, create a button for each program , based on the button, clicked the corresponding program has to be executed and should display the logics for each button includes finding an even or odd number , prime or not , factorial of a number , Armstrong or not use appropriate design to display the output.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      margin: 0;
      font-family: "Segoe UI", Arial, sans-serif;
      background: linear-gradient(135deg, #2c2c2c, #3a3a3a);
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
    }

    .container {
```

```css
  background: #ffffff;
  border-radius: 16px;
  box-shadow: 0 10px 30px rgba(0,0,0,0.25);
  padding: 40px;
  width: 380px;
  text-align: center;
}

h2 {
  margin-bottom: 25px;
  color: #222;
  font-weight: 600;
  letter-spacing: 1px;
}

input {
  width: 100%;
  padding: 12px;
  font-size: 16px;
  border: 1.5px solid #ccc;
  border-radius: 10px;
  margin-bottom: 20px;
  text-align: center;
  transition: all 0.3s ease;
```

```
}

input:focus {
  outline: none;
  border-color: #888;
  box-shadow: 0 0 8px rgba(0,0,0,0.1);
}

.buttons {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 12px;
  margin-bottom: 24px;
}

button {
  padding: 12px;
  font-size: 14px;
  border: none;
  border-radius: 10px;
  cursor: pointer;
  color: #fff;
  font-weight: 500;
  background: linear-gradient(135deg, #5a5a5a, #2f2f2f);
```

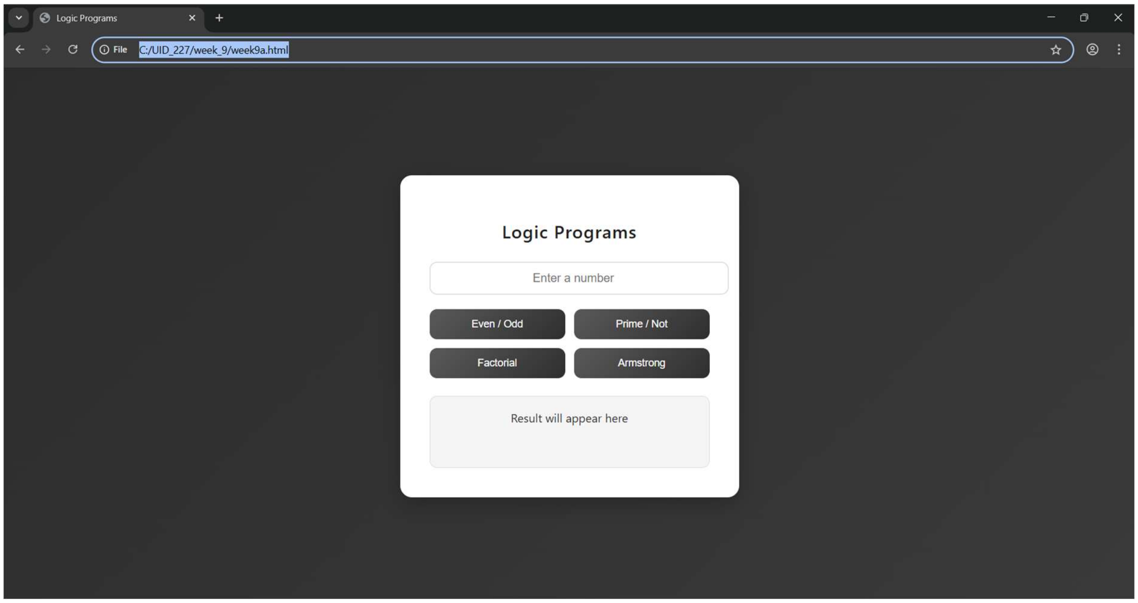
```
      transition: all 0.25s ease;
    }

    button:hover {
      transform: translateY(-2px);
      box-shadow: 0 6px 12px rgba(0,0,0,0.2);
    }

    .output {
      background: #f5f5f5;
      padding: 18px;
      border-radius: 10px;
      min-height: 60px;
      color: #333;
      font-size: 16px;
      border: 1px solid #ddd;
    }
  </style>
</head>

<body>

<div class="container">
  <h2>Logic Programs</h2>
```

```
  <input type="number" id="numInput" placeholder="Enter a
number" />

  <div class="buttons">
    <button onclick="checkEvenOdd()">Even / Odd</button>
    <button onclick="checkPrime()">Prime / Not</button>
    <button onclick="calcFactorial()">Factorial</button>
    <button onclick="checkArmstrong()">Armstrong</button>
  </div>

  <div class="output" id="output">Result will appear here</div>
</div>

<script>
function getInput() {
  const val = document.getElementById('numInput').value.trim();
  if (val === '' || isNaN(val)) {
    document.getElementById('output').textContent = 'Please
enter a valid number.';
    return null;
  }
  return parseInt(val);
}
```

```
function setOutput(text) {
  document.getElementById('output').textContent = text;
}

function checkEvenOdd() {
  const n = getInput();
  if (n === null) return;
  setOutput(n + ' is ' + (n % 2 === 0 ? 'Even' : 'Odd'));
}

function checkPrime() {
  const n = getInput();
  if (n === null) return;
  if (n < 2) { setOutput(n + ' is Not Prime'); return; }
  for (let i = 2; i <= Math.sqrt(n); i++) {
    if (n % i === 0) {
      setOutput(n + ' is Not Prime');
      return;
    }
  }
  setOutput(n + ' is Prime');
}
```

```
function calcFactorial() {
  const n = getInput();
  if (n === null) return;
  if (n < 0) {
    setOutput('Factorial not defined for negative numbers');
    return;
  }
  let result = 1;
  for (let i = 2; i <= n; i++) result *= i;
  setOutput(n + '! = ' + result);
}
function checkArmstrong() {
  const n = getInput();
  if (n === null) return;
  const digits = String(n).split('');
  const power = digits.length;
  const sum = digits.reduce((acc, d) => acc +
Math.pow(Number(d), power), 0);
  setOutput(n + (sum === n ? ' is an Armstrong number' : ' is Not
an Armstrong number'));
}
</script>
</body>
</html>
```

OutPut;



Explaination ;

<!DOCTYPE html> : this tells the browser "yo, this is an HTML5 page", so it knows how to properly load everything

<html> : this is the main root element — everything inside the webpage sits inside this

<head> : contains background stuff (metadata) like title, styles, etc. — not directly visible on screen

<title> : sets the name of the webpage (the tab name you see in browser)

<body> : this is where all visible stuff comes — text, images, buttons, links, tables, etc.

<h> : used for headings (like titles or section names on the page)

<div> : acts like a container/box — helps group things so styling and layout becomes easier

<style> : used to write CSS — basically controls the look (colors, spacing, layout, design)

body (CSS selector) : used to style the full page — here it sets background and centers content

.container : styles the main card — gives white background, padding, rounded edges and shadow

input : styles the input field — controls size, spacing, border and text alignment

button : styles buttons — sets color, size, border and adds hover effect

.output : styles the result box — where output text will be shown

<input> : used to take user input — type="number" means only numbers allowed, id helps JS access it

<div class="buttons"> : groups all buttons together for layout control

<button> : clickable element — onclick triggers JavaScript functions

<div id="output"> : displays the result — initially shows default message

<script> : used to write JavaScript — adds logic and interactivity to the page

getElementById() : used to select an element using its id

.value : gets the value entered in input field

.textContent : used to update or display text inside an element

function : defines a block of code that runs when called

checkEvenOdd() : checks if number is even or odd using modulus % 2 checkPrime() : checks if number is prime using loop and division logic

calcFactorial() : calculates factorial by multiplying numbers from 1 to n

checkArmstrong() : checks if number is Armstrong (sum of digits^power equals original number)

Math.sqrt() : returns square root (used to optimize prime checking)

parseInt() : converts input (string) into integer

isNaN() : checks whether the value is not a number

Week-10

10. Aim: write JavaScript conditional statement to find the sign of product of three numbers using functions .Display an alert box with specified sign .write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result.write a java script conditional statement to find largest of find largest of five numbers using function and onclick event .display an alert box to show the result .write a java script for loop that will iterate from 0 to 15. For each iteration ,it will check if the current number odd or even , and display a message on screen .write a java script program to read 6 subjects marks from the student then computes the average marks of students then , this average is used to determine the corresponding grade.

Program :

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      margin: 0;
      font-family: "Segoe UI", Arial, sans-serif;
      display: flex;
      height: 100vh;
```

```
    background: #121212;
    color: #fff;
}

.left {
    width: 30%;
    background: #1e1e1e;
    padding: 25px;
    border-right: 1px solid #333;
    box-shadow: inset -5px 0 10px rgba(0,0,0,0.4);
}

.right {
    width: 70%;
    background: #181818;
    padding: 30px;
}

h3 {
    margin-top: 0;
    font-weight: 500;
    color: #e0e0e0;
```

```css
}

input {
  width: 95%;
  padding: 12px;
  margin-bottom: 15px;
  border-radius: 8px;
  border: 1px solid #444;
  background: #2a2a2a;
  color: white;
  font-size: 14px;
}

input:focus {
  outline: none;
  border-color: #888;
  box-shadow: 0 0 6px rgba(255,255,255,0.1);
}

button {
  width: 100%;
  padding: 12px;
```

```
    margin-bottom: 12px;
    background: linear-gradient(135deg, #3a3a3a,
#2a2a2a);
    border: 1px solid #444;
    border-radius: 8px;
    color: #fff;
    font-weight: 500;
    cursor: pointer;
    transition: all 0.2s ease;
  }

  button:hover {
    transform: translateY(-2px);
    background: #444;
    box-shadow: 0 4px 10px rgba(0,0,0,0.4);
  }

  #output {
    background: #222;
    padding: 25px;
    border-radius: 10px;
    min-height: 220px;
```

```
        white-space: pre-line;
        border: 1px solid #333;
        font-size: 15px;
        color: #e6e6e6;
      }
    </style>
</head>

<body>

<div class="left">
    <h3>Input Panel</h3>
    <input type="text" id="numbers" placeholder="Enter
values (e.g. 1,2,3)">

    <button onclick="signProduct()">Sign of
Product</button>
    <button onclick="sortNumbers()">Sort
Numbers</button>
    <button onclick="largestNumber()">Largest
Number</button>
    <button onclick="oddEvenLoop()">Odd / Even</button>
```

```
    <button onclick="studentGrade()">Student
Grade</button>
</div>

<div class="right">
    <h3>Output Console</h3>
    <div id="output">Result will appear here</div>
</div>

<script>
function getNumbers() {
    let input = document.getElementById("numbers").value;
    return input.split(",").map(Number);
}

function signProduct() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
        "Enter 3 numbers like: 0,-1,4";
        return;
    }
```

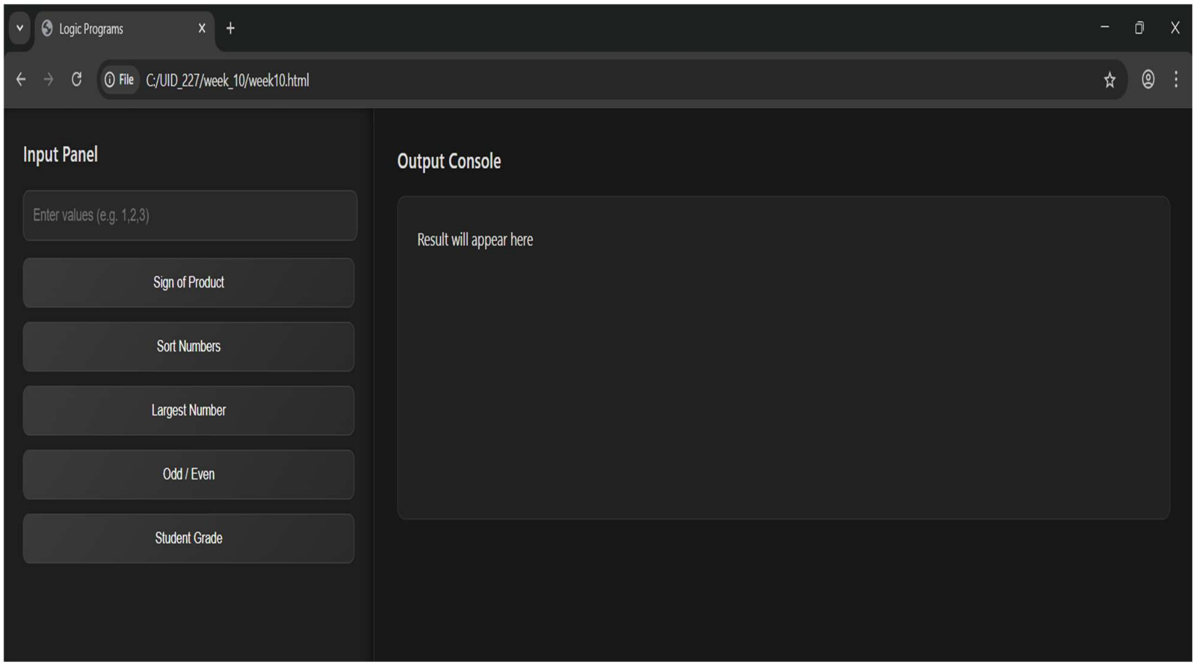
```
  let nums = getNumbers();
  let negative = 0;
  for (let n of nums) {
    if (n < 0) negative++;
  }
  let result = (negative % 2 == 0) ? "+1" : "-1";
  alert(result);
  document.getElementById("output").textContent =
"Sign: " + result;
}

function sortNumbers() {
  let input = document.getElementById("numbers").value;
  if (input === "") {
    document.getElementById("output").textContent =
    "Enter 3 numbers like: 0,-1,4";
    return;
  }
  let nums = getNumbers();
  nums.sort((a, b) => b - a);
  alert(nums.join(", "));
  document.getElementById("output").textContent =
```

```
    "Sorted: " + nums.join(", ");
}

function largestNumber() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
        "Enter 5 numbers like: -5,-2,-6,0,-1";
        return;
    }
    let nums = getNumbers();
    let max = nums[0];
    for (let n of nums) {
        if (n > max) max = n;
    }
    alert("Largest: " + max);
    document.getElementById("output").textContent =
    "Largest: " + max;
}

function oddEvenLoop() {
    let input = document.getElementById("numbers").value;
```

```
  if (input === "") {
    document.getElementById("output").textContent =
    "Enter a number like: 15";
    return;
  }
  let n = Number(input);
  let result = "";
  for (let i = 0; i <= n; i++) {
    if (i % 2 == 0)
      result += i + " is Even\n";
    else
      result += i + " is Odd\n";
  }

  document.getElementById("output").textContent =
result;
}

function studentGrade() {
  let input = document.getElementById("numbers").value;
  if (input === "") {
    document.getElementById("output").textContent =
```

```
        "Enter marks like: 70,80,90,60,75,85";
        return;
    }
    let nums = getNumbers();
    let sum = 0;
    for (let n of nums) sum += n;
    let avg = sum / nums.length;
    let grade;

    if (avg >= 90) grade = "A";
    else if (avg >= 75) grade = "B";
    else if (avg >= 60) grade = "C";
    else grade = "D";

    alert("Average: " + avg + " Grade: " + grade);
    document.getElementById("output").textContent =
    "Average: " + avg + " | Grade: " + grade;
}
</script>

</body>
</html>
```

OutPut;



Explaination ;

<!DOCTYPE html> : tells the browser that this file is written in HTML5 so it should render it properly

<html> : this is the root container — everything in the webpage is placed inside this

<head> : contains background info like title and styling (not shown directly on screen)

<title> : sets the name that appears on the browser tab

<style> : used to add CSS — controls how the page looks (colors, layout, spacing, etc.)

body : styles the whole page — here it sets font, uses flex layout and full screen height

.left : designs the left panel — gives width, background color, padding and border

.right : styles the right section — used to display output with background and spacing

input : used to design the text field — controls width, padding and spacing

button : styles all buttons — sets size, color, removes border and makes it clickable

#output : styles the output area — gives background, padding and keeps line breaks visible

<body> : contains everything visible like input box, buttons and output section

<div class="left"> : left side section where user enters input and clicks buttons

<div class="right"> : right side section where results are displayed

<input> : takes user input — here it accepts text like numbers separated by commas

<button> : used for actions — onclick calls specific JavaScript functions

<div id="output"> : area where output is shown — default text appears before execution

<script> : used to write JavaScript — adds functionality to the webpage

getNumbers() : takes input, splits values by comma and converts them into numbers getElementById() : selects an element using its id

.value : gets the text entered by user

.textContent : used to display or update text inside output box

signProduct() : checks sign of product based on count of negative numbers

sortNumbers() : sorts numbers in descending order using sort function

largestNumber() : finds the biggest number from given list using loop

oddEvenLoop() : prints whether numbers from 0 to given value are odd or even

studentGrade() : calculates average marks and assigns grade based on conditions

split() : breaks string into parts using comma

map(Number) : converts string values into numbers

sort() : used to arrange numbers in order

for...of loop : used to iterate through array elements

if-else : used for decision making

alert() : shows popup message with result

Number() : converts input into number type

Week-11

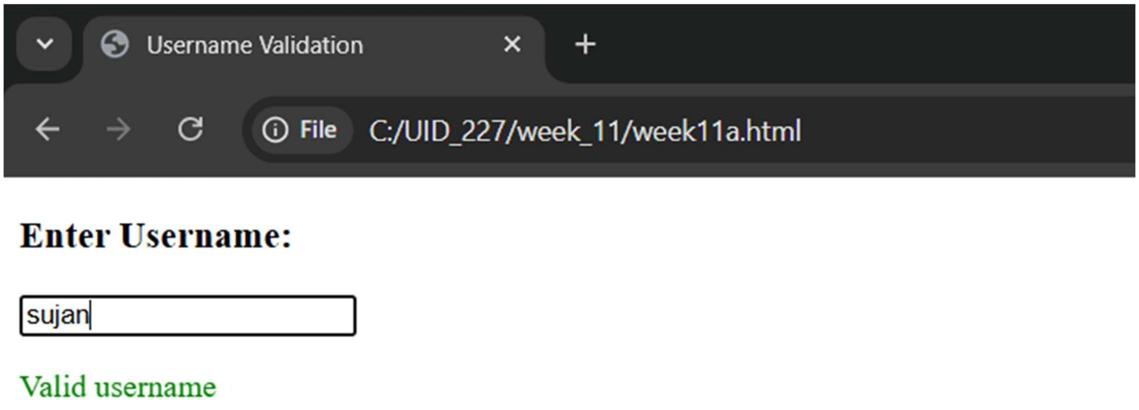
A) Aim: A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Username Validation</title>
</head>
<body>

<h3>Enter Username:</h3>
<input type="text" id="username"
onkeyup="checkUsername()">
<p id="msg"></p>

<script>
function checkUsername() {
```

```
  let username =
document.getElementById("username").value;
  let msg = document.getElementById("msg");

  if (username.length < 5) {
    msg.innerHTML = "Username must be at least
5 characters";
    msg.style.color = "red";
  } else {
    msg.innerHTML = "Valid username";
    msg.style.color = "green";
  }
}
</script>

</body>
</html>
```

Output;



Explanation:

1. document.getElementById("...").value: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. .innerHTML: Changes the actual text inside the <p> tag.

3. <p id="msg"></p>: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Minimum 5 letters required”.

4. <input type="text">: Creates a standard text box for user input.

5. id="username": This is a unique identifier. It acts like a "hook" that allows JavaScript to find this specific input box among all other elements on the page.

6. onkeyup="checkUsername()":Every time the user releases a key (after pressing it), the checkUsername() function is triggered.

7. .length: This property counts the number of characters in the string.

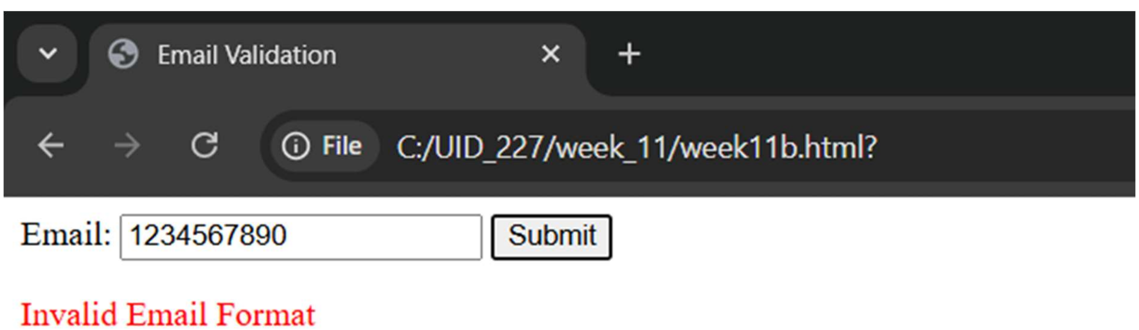
B) Aim: A registration form should validate email format before submission. If invalid, the form should not be submitted.

Program:

```
<!DOCTYPE html>

<html>

<head>

  <title>Email Validation</title>

</head>

<body>


<form onsubmit="return validateEmail()">

  Email: <input type="text" id="email">

  <input type="submit" value="Submit">

</form>


<p id="msg"></p>


<script>

function validateEmail() {

  let email = document.getElementById("email").value;

  let msg = document.getElementById("msg");
```

```
    let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (!email.match(pattern)) {
      msg.innerHTML = "Invalid Email Format";
      msg.style.color = "red";
      return false; // stop submission
    }
    return true;
  }
</script>

</body>
</html>
```

Output:



Explanation:

1. document.getElementById("...").value: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. .innerHTML: Changes the actual text inside the <p> tag.

3. <p id="msg"></p>: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Invalid Email."

4. <input type="text" id="mail" placeholder="Enter email">:The placeholder attribute provides a faint hint inside the box that disappears once the user starts typing.

5. ^[a-zA-Z0-9._%+-]+: Matches the username part before the @. It allows letters, numbers, dots, underscores, and plus signs.

6. @[a-zA-Z0-9.-]+: Matches the domain name.

7. \.[a-zA-Z]{2,4}$: Matches the extension (like .com or .in). The {2,4} means the extension must be between 2 and 4 characters long.

8. document.getElementById("msg").style.color : It gives the user immediate visual feedback: Red for a mistake and Green for success.

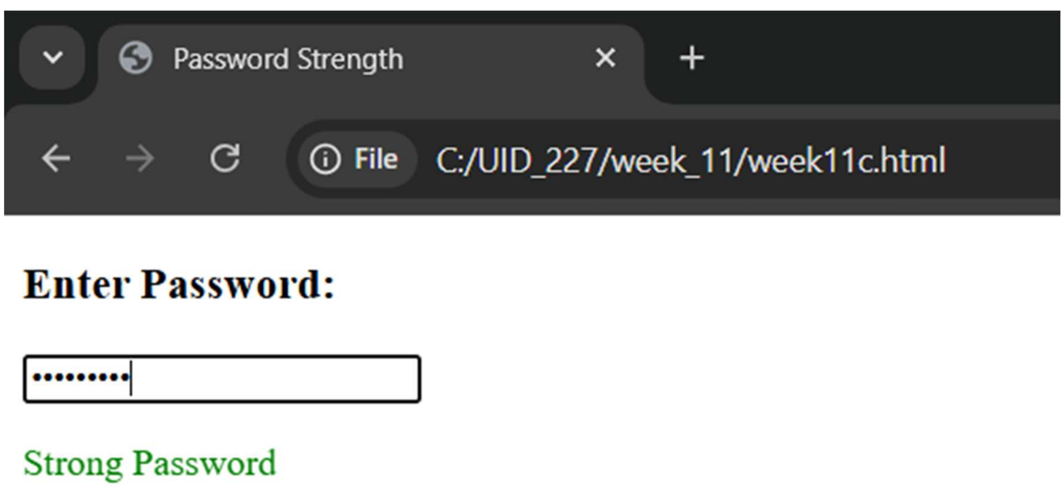
C) Aim: A system should check password strength while typing and display whether it is weak or strong.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Password Strength</title>
</head>
<body>


<h3>Enter Password:</h3>
<input type="password" id="password"
onkeyup="checkPassword()">
<p id="msg"></p>


<script>
function checkPassword() {
  let pass = document.getElementById("password").value;
  let msg = document.getElementById("msg");


  if (pass.length < 6) {
    msg.innerHTML = "Weak Password";
    msg.style.color = "red";
  } else {
```

```
        msg.innerHTML = "Strong Password";
        msg.style.color = "green";
    }
}
</script>

</body>
</html>
```

<u>Output:</u>



Explanation:

1. document.getElementById("...").value: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. .innerHTML: Changes the actual text inside the <p> tag.

3. onkeyup: Fires when the user releases the key.

4. .length: This is a built-in JavaScript property that counts how many characters are in the string.

5. document.getElementById("msg").style.color : It gives the user immediate visual feedback: Red for a mistake and Green for success.

6. <input type="password">: Creates a text box where characters are masked.

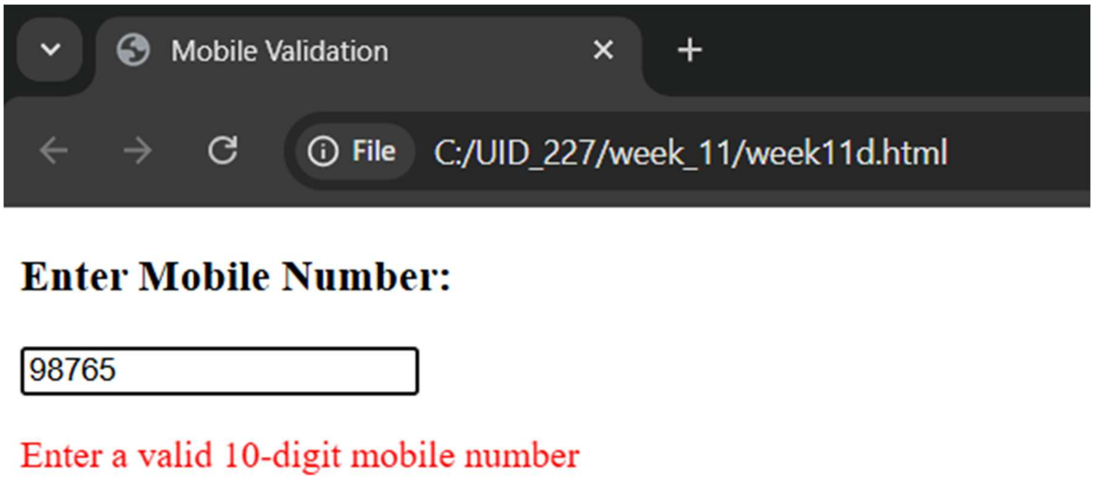
D) Aim: A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mobile Validation</title>
</head>
<body>
```

```html
<h3>Enter Mobile Number:</h3>
<input type="text" id="mobile" onblur="validateMobile()">
<p id="msg"></p>

<script>
function validateMobile() {
  let mobile = document.getElementById("mobile").value;
  let msg = document.getElementById("msg");

  let pattern = /^[0-9]{10}$/;

  if (!mobile.match(pattern)) {
    msg.innerHTML = "Enter a valid 10-digit mobile number";
    msg.style.color = "red";
  } else {
    msg.innerHTML = "Valid mobile number";
    msg.style.color = "green";
  }
}
</script>

</body>
</html>
```

Output:



Explanation:

1. document.getElementById("...").value: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. .innerHTML: Changes the actual text inside the <p> tag.

3. let pattern = /^[0-9]{10}$/; : This is a regular expression.

· ^ : Start of the string.

· [0-9] : Only digits are allowed.

· {10} : Exactly 10 digits required.

· $: End of the string.

4. document.getElementById("msg").style.color : It gives the user immediate visual feedback: Red for a mistake and Green for success.

5. oninput="Mobilenumber()":fires every single time the value inside the box changes.

6. .match(pattern) : Compares the user's input against the Regex.

E) Aim: 5. Design an HTML page, such that the registration form must validate: Name (not empty), Email (valid format),Password (minimum 6 characters).

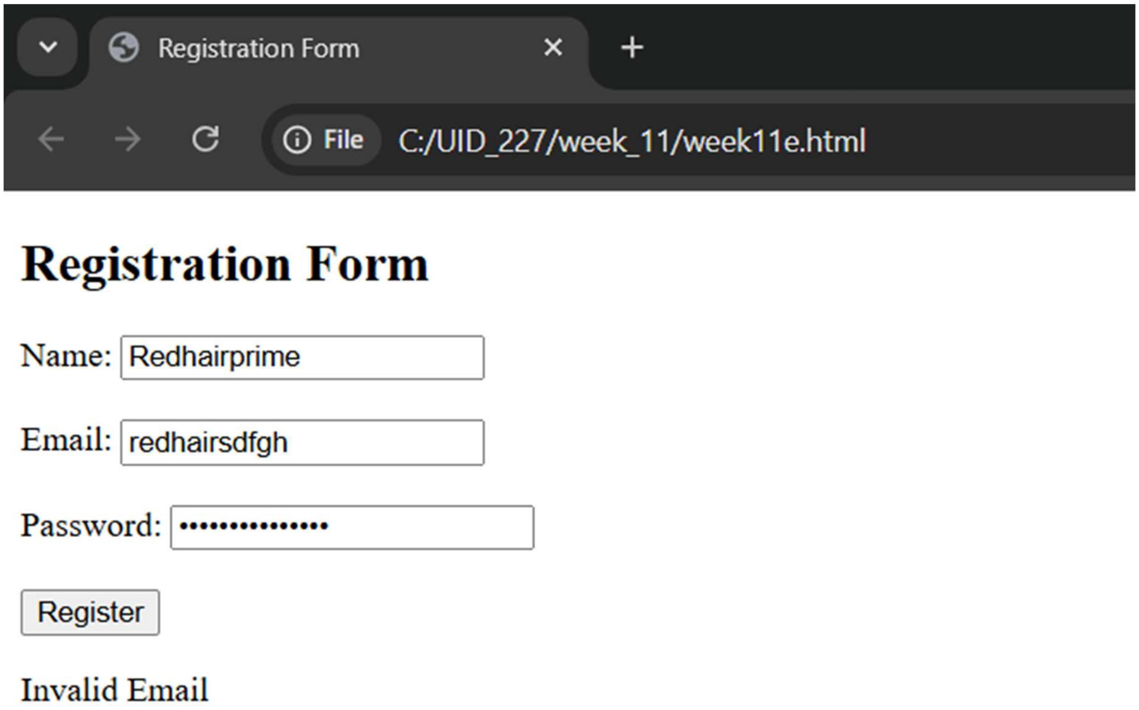
Program:

```
<!DOCTYPE html>

<html>

<head>

   <title>Registration Form</title>

</head>

<body>


<h2>Registration Form</h2>


<form onsubmit="return validateForm()">
```

```
  Name: <input type="text" id="name"><br><br>

  Email: <input type="text" id="email"><br><br>

  Password: <input type="password" id="password"><br><br>

  <input type="submit" value="Register">

</form>


<p id="msg"></p>


<script>

function validateForm() {

  let name = document.getElementById("name").value;

  let email = document.getElementById("email").value;

  let password =
document.getElementById("password").value;

  let msg = document.getElementById("msg");


  let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;


  if (name === "") {

    msg.innerHTML = "Name cannot be empty";

    return false;

  }


  if (!email.match(emailPattern)) {
```

```
      msg.innerHTML = "Invalid Email";
      return false;
   }

   if (password.length < 6) {
      msg.innerHTML = "Password must be at least 6
characters";
      return false;
   }

   return true;
}
</script>

</body>
</html>
```

Output:



<u>Explanation:</u>

1. document.getElementById("...").value: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. return false: Returning false stops the form from submitting.

3. return true: Returning true means the data is valid and the form can proceed.

4. <form>: Defines a container for user input.

5. <input>:

· type="text": Creates a basic single-line text box.

· type="password": Creates a text box where characters are masked.

· type="submit": Creates a button that automatically triggers the form's submission.

· id="...": A unique identifier for the element, which the JavaScript uses to "grab" the data entered by the user.

6. <p id="msg"></p>: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Invalid Email."

Week-12

Aim: To install and setup the Electron Framework application.
Procedure:

Step 1: Go to the web browser and download node.js.

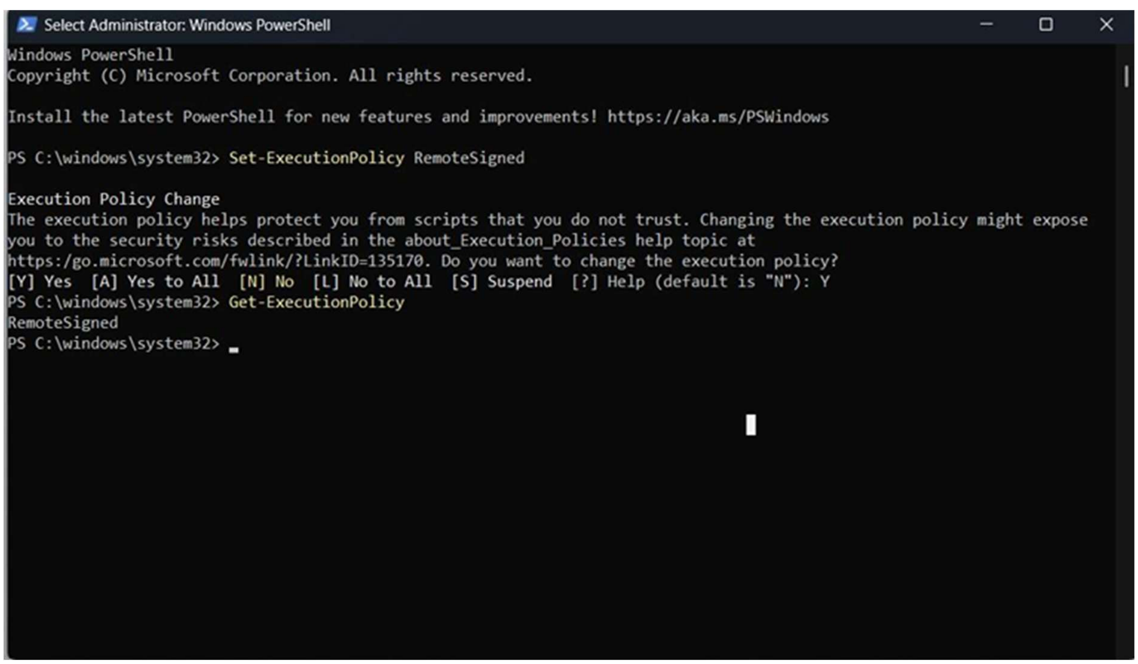
Step 2: After installing the Node.js, complete the setup process.

Step 3: After installing node.js, Open the command prompt and navigate to the folder where your project is located.

Step 4: And give the prompt “node –v” and also "npm –v".

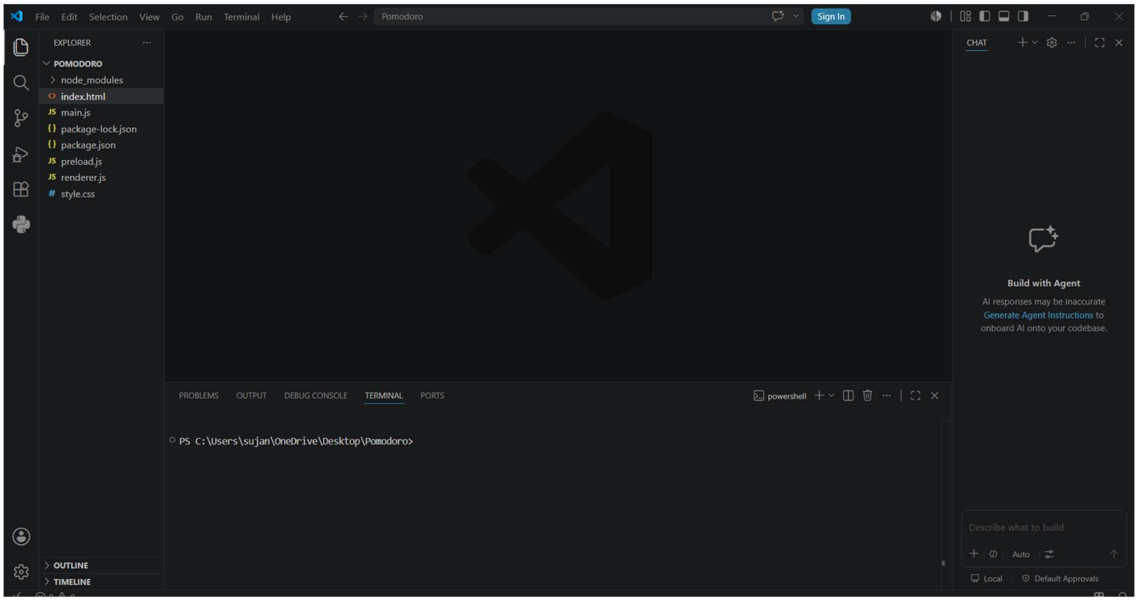
Step 5: If you get the versions then it is successfully installed.

Step 6: Now open PowerShell as Administrator and give the prompt "Set-ExecutionPolicy RemoteSigned".



Step 7:

Create a new empty folder in VS code.



An empty folder is created with the name as Electron Project in Visual Studio.

Step – 8:

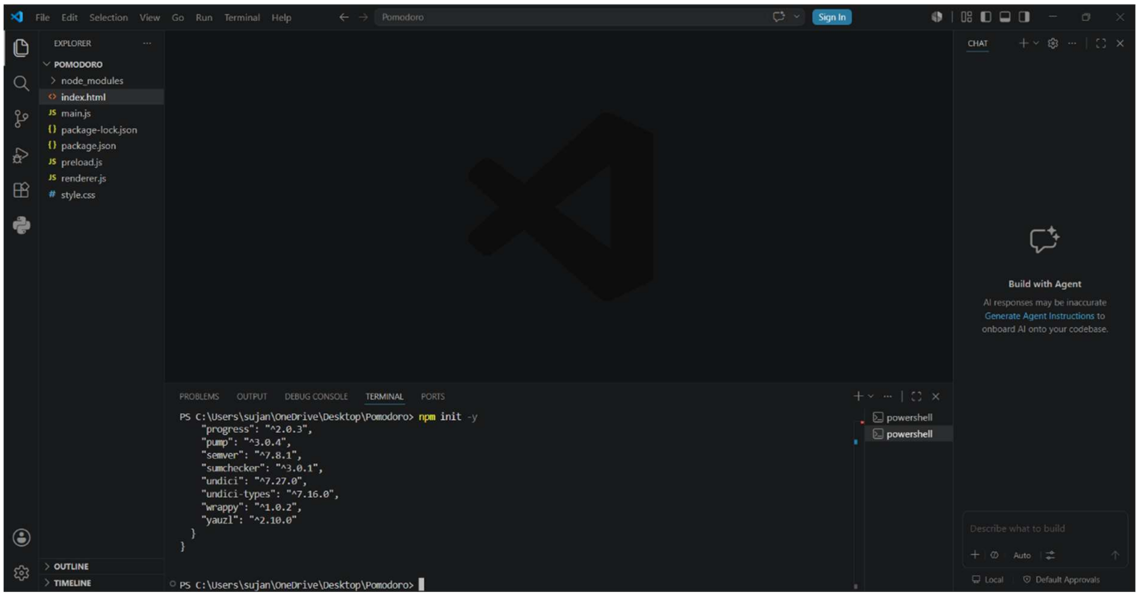
Open Command prompt in terminal.

Step – 9:

Type the below command in the terminal section.

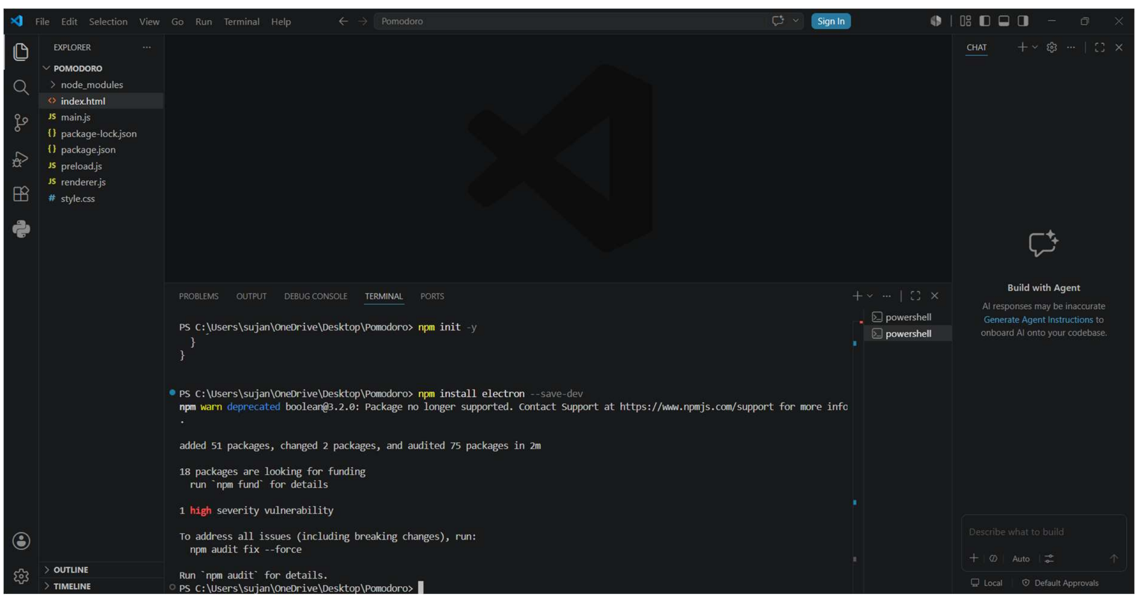
By giving the command, the below information will be displayed

And also “package.json” named file will be displayed in the folder section

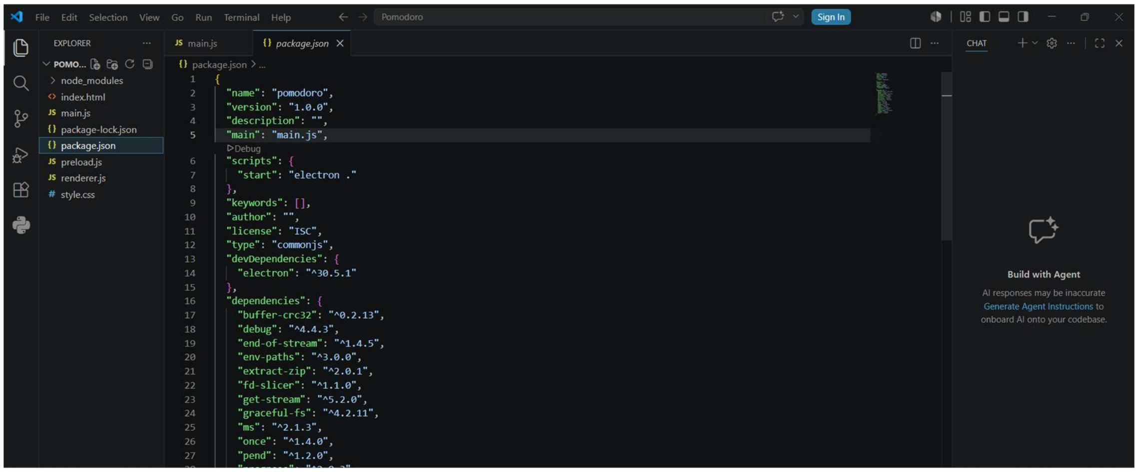


Step – 10:

Type this command “npm install electron - - save - dev.”

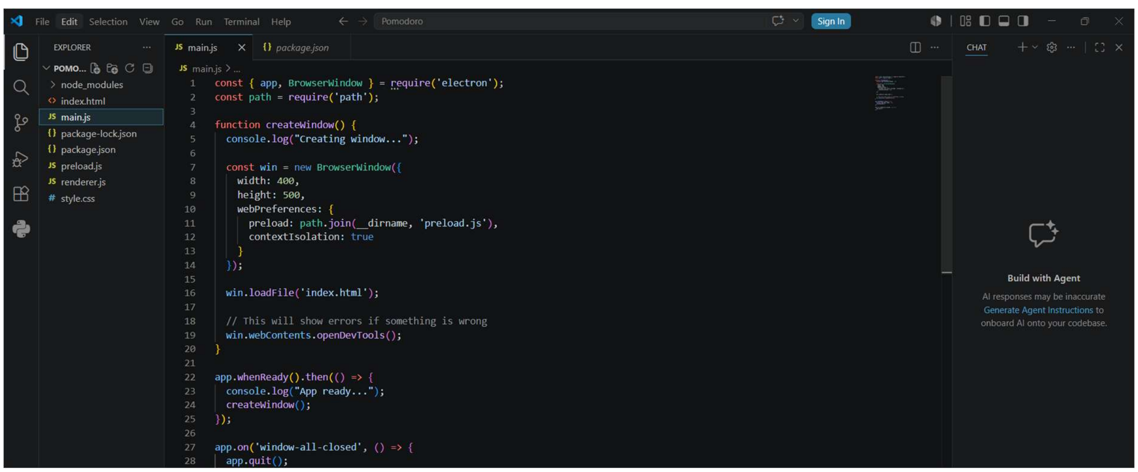


By giving the command, the above information will be displayed and also few more files will be displayed in the file section., Step 11: We added a file named “main.js”,



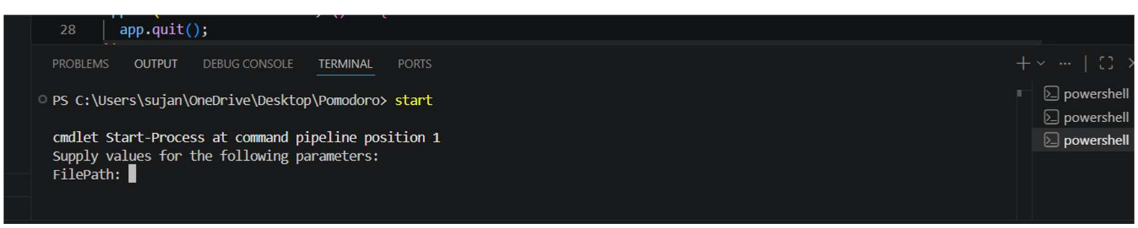
Step – 12:

Make sure the name of the html file in main.js and the html file you added in the folder are same.



Step – 13:

Give start prompt.



Step-14:

The OutPut will be displayed.